The Role of Self-Selection in Agricultural Productivity Gap (APG): Evidence from Indonesia

Isabella Germinario

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Abstract

This paper examines the contribution of worker self-selection to Indonesia's agricultural productivity gap (APG), using the first three waves of the Indonesia Family Life Survey (1993–2000). It adopts a correlated random coefficient (CRC) framework that recovers unobserved comparative advantage from individuals' sector-choice trajectories without imposing a parametric distribution on latent abilities and distinguishes selection-relevant comparative advantage from other time-invariant worker heterogeneity. The estimated selection effect is -0.05, with a standard error of 0.38, suggesting, albeit imprecisely, that unobserved comparative advantage generates little differential payoff across sectors. The average earnings gap between non-agricultural and agricultural workers is approximately 0.95 log points. Under the normalization adopted in the analysis, sector-wide efficiency differences account for about 37% of the gap, observed worker characteristics for about 39%, and unobserved worker heterogeneity for the remaining 24%. Within this unobserved component, selection on unobserved comparative advantage accounts for only about -0.5% of the APG, weakly narrowing rather than widening it. Thus, although workers differ substantially across sectors, sorting on unobserved comparative advantage plays only a minor role in explaining the APG. The observed gap therefore cannot be interpreted directly as the aggregate productivity gain available from reallocating existing workers out of agriculture.

1 Introduction

The Agricultural Productivity Gap (APG) is defined as the ratio of value added per worker in non-agriculture to that in agriculture, and is used to quantify the sectoral productivity gap (Gollin et al., 2014). This term is widely adopted in the structural transformation literature, which was pioneered by Lewis' seminal analysis (1954) of the dual economy and inter-sectoral labour movements. In what follows, I use the terms "sectoral productivity gap" and "APG" interchangeably.

In poor countries, agriculture employs the majority of the labour force (Gollin et al., 2002; Caselli, 2005; Chanda and Dalgaard, 2008; Restuccia et al., 2008). In most of those low-income countries, value added per worker in non-agriculture is several times that in agriculture (Gollin et al., 2014; Lagakos and Waugh, 2013; McMillan and Rodrik, 2014). Taken at face value, the presence of a large APG where a high share of the workforce remains in agriculture resembles an unexploited arbitrage: aggregate output per worker could seemingly rise simply by reallocating labour out of agriculture.

Whether the arbitrage is real, however, is exactly what is in question, and depends on the mechanism behind this gap. In the literature, two prominent potential explanations for substantial APG in developing countries are: One is misallocation, where distortions and frictions that keep land, capital, inputs, and labour from flowing to their highest-productivity uses—both within agriculture and across sectors (Restuccia et al., 2008; Bryan et al., 2014; Munshi and Rosenzweig, 2016; Alvarez-Cuadrado et al., 2017; Pulido and Świecki, 2019; Lagakos, 2020; Gai et al., 2021). The alternative hypothesis emphasizes self-selection: a subsistence food requirement keeps workers with low agricultural productivity in farming, depressing agriculture's average productivity and, by composition, widening the observed cross-sector productivity gap in poorer countries (Lagakos and Waugh, 2013; Herrendorf and Schoellman, 2018; Alvarez, 2020; Hamory et al., 2021).

While both explanations can, in theory, generate significant agricultural productivity gaps, they differ in mechanisms and binding constraints; therefore, they require different policy levers. If misallocation dominates, the blocked arbitrage can be released by dismantling barriers to mobility, reallocating labour relatively quickly. If selection dominates, much of the gap reflects who is in each sector rather than a difference the marginal mover would realize, so the apparent arbitrage is smaller than the raw gap suggests; raising aggregate productivity then depends less on moving workers than on raising the productivity in both sectors—through human capital and technology—a slower undertaking. Distinguishing between these forces is crucial for understanding what drives observed gaps and how policy should respond.

In Indonesia, the components of the APG shift with development—the composition of value added moves toward non-agriculture while agriculture's relative share of employment falls—yet these movements largely offset each other, so the APG itself is largely sustained, fluctuating rather than closing over the period (Gollin et al., 2025). What sustains that persistent gap—frictions or selection—is the question that requires further investigation.

This paper examines the extent to which individual selection based on unobserved comparative advantage contributes to sectoral productivity gaps in Indonesia, a country that moved from low- to a low-middle-income status and underwent considerable structural transformation between 1993 and 2014. The answer to this question has far-reaching policy implications. Understanding how much individual sorting contributes to APG is therefore central to designing effective strategies for structural transformation and sustainable growth in low-income countries.

Since Lagakos and Waugh's (2013) self-selection hypothesis, a growing literature has sought to measure how much selection explains productivity gaps in developing countries (Lagakos and Waugh, 2013; Pulido and Święcki, 2019; Alvarez, 2020; Lagakos et al., 2020; Alvarez-Cuadrado et al., 2020; Adamopoulos et al., 2022). These studies consistently report that selection is important, but the estimated magnitudes vary widely. This variation reflects the difficulty of the task: selection stems from latent, sector-specific heterogeneity that workers internalize as they choose sectors. These unobserved abilities affect both sector choice and earnings, inducing endogeneity. Moreover, the data record only the chosen-sector outcomes—not the counterfactual—making it difficult to disentangle selection effects. The stakes are high, because mismeasuring selection risks misdirecting policy—leading governments to focus on moving workers across sectors when the real constraint may be technology, or vice versa.

In the APG literature, there are two prevailing approaches. The first applies two-way fixed effects (TWFE) to panel data, using workers who switch sectors to estimate the within-individual change in earnings associated with sectoral mobility while absorbing time-invariant individual heterogeneity. This approach has been interpreted either as evidence of low mobility barriers, when switcher gains are small, or as evidence of selection, by comparing sectoral earnings gaps before and after including individual fixed effects. However, TWFE is not designed to recover selection on unobserved comparative advantage. A fixed-effect comparison captures time-invariant unobserved differences across actual workers or within actual switchers. Selection on unobserved comparative advantage, by contrast, is a counterfactual concept: it concerns whether the same selected workers receive differential rewards from their unobserved ability across the two potential sectoral earnings regimes. For example, for non-agricultural workers, the selection effect measures how much more or less their unobserved ability is rewarded in non-agriculture than it would have been had they worked in agriculture. TWFE does not directly recover this counterfactual comparison.

The second approach imposes a distribution for unobserved abilities (e.g., joint normal or Fréchet) and recovers a closed-form parameter in the spirit of the Roy (1951) framework. While tractable, these strong assumptions rarely have empirical support (Heckman and Honore, 1990) and can heavily influence estimated magnitudes. These challenges motivate an alternative approach that targets unobserved comparative advantage directly while avoiding restrictive distributional assumptions.

To address these limitations, this paper adopts a Correlated Random Coefficient (CRC) framework, following Suri (2011), and applies it to the context of sectoral choice and productivity gaps. The CRC approach models unobserved abilities as time-invariant and sector-specific, distinguishing individual fixed effects by directly modelling unobserved comparative advantages at the individual-sector level and enabling the separation of sector selection-relevant abilities from those that are not. By exploiting the information embedded in individuals' sectoral choice histories, this method reveals their latent comparative advantages and recovers the selection effect without imposing distributional assumptions on unobserved heterogeneity. Moreover, the CRC selection term admits a transparent interpretation in the language of the classic Roy model, which the next chapter develops in full.

I depart from Suri's original framework in two ways. First, I adapt the model to the sectoral productivity setting by redefining absolute advantage to reflect its relationship to comparative advantage and sector choice, rather than treating it as independent of the sorting mechanism as in the original framework. Second, after recovering the selection parameter, I add an aggregation step that links the individual-level CRC estimates to the observed agricultural productivity gap (APG). This step decomposes the APG into sector-wide productivity differences, observed worker characteristics, and the contribution of unobserved comparative advantage. This augmentation is not part of the original framework, but it is necessary for this paper because the goal is not only to estimate the selection effect, but also to quantify how much it accounts for the observed APG.

In addition to these modifications, I provide an economic interpretation of the identification strategy. Sectoral choice histories are treated as revealed signals of unobserved comparative advantage: workers' observed trajectories across agriculture and non-agriculture contain information about their latent sector-specific abilities. This interpretation clarifies why the CRC framework can recover selection-relevant heterogeneity from panel choice histories rather than from parametric assumptions about the distribution of latent abilities.

To examine the role of individual sorting on the sectoral productivity gap, I use the Indonesia Family Life Survey (IFLS) (Frankerberg et al., 1995), a nationally representative panel dataset spanning five waves from 1993 to 2014 and covering about 80 percent of the population. During this period, Indonesia moved from low- to lower-middle-income status, with per capita incomes roughly doubling and the share of agricultural employment falling from approximately 46% in 1993 to 34% in 2014 (World Bank Group, 2025). The IFLS is particularly well-suited for studying sorting and productivity gaps because of its rich panel structure, which tracks both sector choices and earnings. In this paper, I focus on the first three waves (1993–2000), before Indonesia's sweeping political and institutional changes after 2000, to provide a clean setting for analyzing sectoral productivity gaps.

Applying this framework to the first three waves of the IFLS (1993–2000), I estimate an average earnings gap of about 0.95 log points between non-agricultural and agricultural workers. Individual sorting on unobserved comparative advantage plays only a limited role in explaining this gap. Roughly 80% of workers remain in their initial sector, while the remaining workers switch sectors at least once over the sample period. The estimated selection effect is about -0.05, with a standard error of approximately 0.38, and is small and statistically insignificant.

In the APG decomposition, sector-wide productivity differences account for nearly 37% of the observed gap, while observed characteristics account for about 39%. The remaining unobserved component accounts for about 24%, but this should not be interpreted as the selection effect itself. Under a normalization that measures unobserved comparative advantage relative to its average, the selection term associated with unobserved comparative advantage contributes approximately -0.5% of the APG and therefore works weakly against, rather than widens, the observed gap. Its small magnitude should nevertheless be interpreted cautiously, because the underlying selection parameter is imprecisely estimated.

This contrast arises for two reasons. First, unlike TWFE, which absorbs all time-invariant individual heterogeneity into a single fixed effect, the empirical framework used here separates the unobserved component of the APG from the selection term associated with sorting on unobserved comparative advantage. This distinction shows that the sizable contribution of unobserved heterogeneity—approximately 24% of the APG—should not be interpreted as evidence of a correspondingly large selection effect. Second, the framework avoids imposing a parametric distribution on latent abilities. Instead, it uses individuals' sector-choice trajectories to recover the empirical distribution of unobserved comparative advantage, revealing much greater dispersion than a normal benchmark would imply. Together, these features explain why sorting on unobserved comparative advantage plays only a minor role in this setting.

This paper makes two main contributions. Methodologically, it extends a correlated random coefficient (CRC) framework to study sectoral productivity gaps and estimate selection on unobserved comparative advantage without relying on strong distributional assumptions. The approach aligns the treatment of absolute advantage in the CRC model with the APG literature, uses individuals' sector-choice trajectories to recover unobserved comparative advantage, and aggregates the individual-level selection estimates to decompose the observed sectoral gap. Empirically, the paper applies this framework to Indonesia and helps reconcile conflicting estimates in the APG literature. It shows that larger selection effects reported elsewhere can partly reflect methodological choices embedded in two dominant approaches: two-way fixed effects, which absorb all time-invariant heterogeneity, and parametric selection corrections, which impose distributional structure on latent abilities. Together, these contributions show that sector-wide productivity differences account for a substantial share of the observed gap in Indonesia during the 1990s, while sorting on unobserved comparative advantage plays only a minor role.

The remainder of this paper proceeds as follows. Section 2 reviews the related literature on sectoral productivity gaps and self-selection. Section 3 defines the measurement of APG and develops the empirical framework based on Suri's (2011) correlated random coefficient model, detailing adaptations for this setting. Section 4 sets out the identification strategy and estimation procedures. Section 5 introduces the dataset, provides descriptive evidence, and presents the baseline results for the first three IFLS waves. Section 6 examines the robustness of these results to alternative sample construction and additional controls. Section 7 discusses how the findings relate to the existing TWFE and Roy-model evidence and draws out policy implications. Section 8 concludes.

2 Literature Review

This section situates the paper within the literature on sectoral productivity gaps (APG), with particular attention to studies that examine the role of individual self-selection. Two dominant hypotheses explain persistent productivity gaps in low-income countries: misallocation of resources and sorting based on comparative advantage. While not mutually exclusive, distinguishing their relative importance is critical for interpreting observed gaps. If misallocation dominates, then the APG signals inefficiencies that policy can alleviate. If sorting dominates, much of the gap reflects workers' latent abilities, so reallocating labour across sectors yields less than the raw gap would imply, and the binding constraint shifts to raising workers' own productivity.

The review proceeds in three steps. First, it discusses the origins of the APG literature and the theoretical motivation for focusing on selection. Second, it synthesizes empirical evidence on the magnitude of selection effects, highlighting the wide range of reported estimates. Third, it identifies the limitations of prevailing empirical approaches, motivating the need for the alternative framework developed in this paper.

2.1 APG and the Selection Hypothesis

The study of sectoral productivity gaps is rooted in classic theories of structural transformation and growth (Lewis, 1954; Kuznets, 1971). A consistent empirical finding is that agricultural productivity lags far behind non-agricultural productivity in poor countries, with gaps far larger than those observed in rich economies (Gollin et al., 2002; Restuccia et al., 2008; Gollin et al., 2014; McMillan and Rodrik, 2014). Using national accounts for 151 countries, Gollin et al. (2014) report a mean APG of 3.5 and a median of 2.6, with values ranging from about 1.3 at the 10th percentile of the cross-country distribution to 6.8 at the 90th percentile. The gap is larger, on average, in poorer countries—mean 2.0 in the richest income quartile versus 5.6 in the poorest—but with wide dispersion within income groups, so a country's APG level is only loosely tied to its stage of development.

Early critics questioned whether such large gaps merely reflected measurement error. Yet Gollin et al. (2014) show that, after adjusting for sector differences in hours worked and human capital across 72 countries, the average gap falls by only about a third—to a mean of 2.2 (median 1.9), still ranging from 1.0 to 6.4 across the distribution—indicating that measurement refinements are not first-order. Herrendorf and Schoellman (2018) further show, across forty-two censuses in thirteen countries over seven decades, that poor countries consistently exhibit large wage differences between sectors. Together, these studies confirm that APGs are real, persistent, and central to understanding income disparities, especially in poor countries.

The literature offers two leading explanations for large APGs. The first is misallocation (Restuccia and Rogerson, 2017; Gollin and Kaboski, 2023). It arises when policy and market frictions prevent resources and workers from moving to their most productive uses, keeping economies below their potential frontier (Restuccia et al., 2008). In agriculture, farm-level distortions and land-market failures compress farm size and weaken the link between inputs and output, leading to large productivity losses (Adamopoulos and Restuccia, 2014; Chen et al., 2023). Mobility and institutional barriers also hinder efficient allocations: credit and risk constraints curb migration from rural areas, while social or policy institutions limit occupational mobility (Bryan et al., 2014; Munshi and Rosenzweig, 2016; Pulido and Świecki, 2019; Gai et al., 2021). Finally, differences in capital–labour substitutability across sectors point to underlying technology gaps, suggesting that agricultural productivity could rise substantially if such technologies diffused more widely (Alvarez-Cuadrado et al., 2017; Chen, 2020).

By contrast, selection highlights that individuals choose sectors based on comparative advantage. Building on Roy's (1951) framework, Lagakos and Waugh (2013) formalize this hypothesis by combining comparative advantage with subsistence food requirements: in low-productivity economies, many workers must remain in agriculture, creating large dispersion in farm productivity, whereas in high-productivity economies only highly skilled farmers remain, raising average productivity. When comparative and absolute advantage are positively correlated, sorting amplifies the APG. Later work notes that selection and misallocation likely coexist Lagakos (2020), and that the correlation between comparative and absolute advantage may be weak or even negative in poor countries (Alvarez-Cuadrado et al., 2020).

Thus, the selection hypothesis provides a compelling microfoundation for observed APGs; the remaining challenge is to measure its magnitude—an issue I discuss next.

2.2 Empirical Analysis of Selection on APG

Following the theoretical foundation of the selection hypothesis proposed by Lagakos and Waugh (2013), a growing empirical literature has sought to quantify how individual sorting across sectors contributes to the Agricultural Productivity Gap (APG). Two main approaches have emerged in addressing the challenge of unobserved comparative advantage, which drives sectoral choice. One relies on panel fixed-effects estimators using sector switchers to control for time-invariant unobserved heterogeneity. The other assumes a specific distribution for unobserved sectoral abilities within a structural framework. Across studies, the estimated contribution of selection varies widely—from roughly one-third to over ninety percent—leaving unsettled debate on the relative importance of selection and misallocation.

Structural Approach

A group of studies—Lagakos and Waugh (2013); Pulido and Świecki (2019); Herrendorf and Schoellman (2018); Alvarez (2020)—combine micro evidence with structural models to assess how selection and misallocation contribute to sectoral productivity gaps. These papers converge methodologically, using reduced-form estimates or panel moments as inputs for structural analysis rather than relying solely on macro data.

Some studies assume functional forms for unobserved abilities. Lagakos and Waugh (2013) assume dependent Fréchet-distributed sector-specific latent abilities, while Pulido and Świecki (2019) adopt a joint-normal distribution of unobserved heterogeneity. These assumptions allow analytical tractability and closed-form aggregation but inherently shape the magnitude of selection: the Fréchet distribution's heavy tails mechanically amplify selection effects, producing large apparent heterogeneity even when underlying ability dispersion is moderate. Moreover, as Heckman and Honore (1990) emphasize, there is little empirical evidence that unobserved abilities are normally distributed, raising concerns about the sensitivity of estimates to distributional assumptions.

Acknowledging the limitations of distributional assumptions, Alvarez (2020) uses Brazilian administrative and survey panels (1993–2016) to compare wage changes for sector switchers and multi-sector workers, finding modest within-individual wage gains—9 log points for manufacturing and 4 for services—relative to an average inter-sector gap of 48 log points. Herrendorf and Schoellman (2018) analyze 13 countries' censuses and complement them with panel evidence from the PSID (United States) and other studies for Brazil and Indonesia, concluding that observed mobility frictions are modest relative to ability-based sorting. Although these studies avoid imposing functional forms on latent abilities, they rely on fixed-effects estimates for sector switchers to inform their models. This reliance draws inference from a sub-population of workers with marginal gains relative to non-switchers, limiting the representativeness of estimated selection effects.

Reduced-Form Approaches

Beyond using fixed effects as a microfoundation for structural calibration, some studies apply similar methods directly to estimate selection, exploiting switchers identified in longitudinal microdata. Aiming at directly evaluating the impact of individual sorting on APG, Hamory et al. (2021) employ the Two-Way Fixed Effect (TWFE) estimator in panel data without a macro structural model and interpret the productivity gap across sectors between with and without controlling for the fixed effect as the impact of individuals' sorting on the APG. They use long panels for Kenya and Indonesia and estimate an average earnings gap of roughly 70 log points between agriculture and non-agriculture. Controlling for individual fixed effects reduces this to about 22–24 log points, implying that 67–92 percent of the observed APG reflects individual sorting. This framework assumes that fixed effects fully capture unobserved heterogeneity relevant to sectoral choice. In practice, however, they absorb both general ability and sector-specific traits, yielding a local estimate for marginal movers rather than a population-level measure of comparative advantage.

Related studies extend this framework to migration contexts to recover the selection effect on APG through rural-urban movers. Lagakos et al. (2020), using panel data from six developing countries, find that once worker fixed effects are included, estimated migration costs decline substantially and average returns to migration are around 23% in five of the six countries—suggesting limited misallocation. By contrast, Gai et al. (2021) exploit a natural policy intervention in rural China as an instrument and frame their analysis as an event study. They estimate the local treatment effect (LATE) for the agriculture-to-nonagriculture switchers affected by the policy intervention and find that those sector switchers face migration costs equivalent to approximately 55% of their earnings in the nonagricultural sector. Using a control function approach, they estimate an average treatment effect (ATE) on sectoral earnings in rural China during the study period of 2003-2012 and recover an underlying APG of 46 log points once the selection on unobserved sector-specific abilities is accounted for. Given that the observed APG in their data is 68 log points, they conclude that approximately 32% of the sectoral productivity gaps are explained by individuals' selection into sectors based on their latent skills.

Together, these results contribute to the ongoing debates on the relative roles of selection and misallocation.

Synthesis and limitations

Overall, the literature demonstrates that worker sorting contributes meaningfully to the

APG, yet its precise magnitude remains uncertain. In terms of how unobserved comparative advantage is identified, the two prevailing approaches are (a) panel fixed-effects estimation relying on switchers, and (b) structural modeling based on assumed distributions of latent abilities.

Despite valuable insights, these approaches face consequential limitations in addressing two core challenges when estimating the selection effect on APG: (1) measuring sector-specific heterogeneous abilities; and (2) disentangling endogeneity induced by unobserved comparative advantages. The panel approach that controls for individual fixed effects does not allow an appropriate formulation of comparative advantage, because it treats unobserved heterogeneity as an individual-specific component rather than distinguishing sector-specific abilities for each person. In addition, this approach ignores non-switchers and risks attributing all unobserved abilities to sectoral choice—including traits unrelated to sector selection, such as general ability or work ethic—thereby overstating the selection effect. By contrast, the structural approach based on the generalized Roy model estimates selection by assuming a specific distribution for unobserved sector-specific abilities (e.g., joint normal or Fréchet). However, these functional assumptions lack empirical support and can materially influence the estimated effects.

These limitations leave the magnitude of the selection effect highly sensitive to methodological choices. In practice, existing studies report wide-ranging estimates—significant but inconsistent in size—because empirical strategies often conflate sector-specific comparative advantage with general latent traits that do not govern sector choice, and they lean on restrictive assumptions to handle heterogeneity and endogeneity. To address these challenges, this paper adopts an alternative framework that models sector-specific heterogeneity as individual deviation from sector-average productivity while avoiding parametric distributional assumptions about unobserved heterogeneity. Specifically, I adapt the correlated random-coefficients (CRC) framework of Suri (2011), in the distribution-free spirit of Lemieux (1998), which avoids strong parametric restrictions and better isolates the sector-relevant component of individual abilities. The next section details the implementation and identification of each component.

Appendix 9.1 provides a fuller description of Suri's original CRC framework—and its application to hybrid seed adoption in Kenya—for readers seeking additional background before turning to this paper's adaptation to the APG setting.

3 Empirical Model

This section outlines the empirical model of this paper by extending the approach in Suri (2011) to the APG literature. I begin by modelling individual earnings by using the Mincerian representation for human capital. Building on this foundation, I follow the method in Lemieux (1998) and Suri (2011) to incorporate heterogeneous latent skills across sectors into this model, where absolute and comparative advantages are defined. Next, I put together the main empirical model that enables the measurement of how self-selection affects individual earnings. Finally, I illustrate how to draw the impact of individual selection on the sectoral productivity gap through aggregation.

3.1 Model Setup

As individual sorting is based on comparing the potential earnings in each sector, the starting point is to model how the choice of sector affects an individual's potential earnings. In an economy, an individual i at time t can choose to work in one of two sectors: $j \in \{n, a\}$, where n refers to the non-agricultural sector and a to the agricultural sector. An individual's potential earnings in each sector at each period are determined by the sector productivity, A_t^j, and her own human capital, h_{it}^j, as expressed in equations (1) and (2). In these two equations, P_t^j represents the price level at each sector j.

$$W_{it}^n = P_t^n A_t^n h_{it}^n \tag{1}$$

$$W_{it}^a = P_t^a A_t^a h_{it}^a \tag{2}$$

Following Mincer, the human capital h_{it}^j can be expressed as an exponential function of observed and unobserved characteristics, see equations (3) and (4), where X_{it} is a vector of observed characteristics endowed by individual i at time t, and U_{it}^j is a vector of individual

i's unobservable in sector j at time t.

$$h_{it}^{n} = exp(X_{it}\gamma^{n} + U_{it}^{n}) \tag{3}$$

$$h_{it}^{a} = exp(X_{it}\gamma^{a} + U_{it}^{a}) \tag{4}$$

Substitute equations (3) and (4) into (1) and (2), the individual potential earnings in each sector j and time t can be represented as equations (5) and (6).

$$W_{it}^{n} = P_{t}^{n}A_{t}^{n}exp(X_{it}\gamma^{n} + U_{it}^{n}) \tag{5}$$

$$W_{it}^{a} = P_{t}^{a}A_{t}^{a}exp(X_{it}\gamma^{a} + U_{it}^{a}) \tag{6}$$

Taking logs, I can obtain equations (7) and (8), where the lower cases represent the logarithmic terms.

$$w_{it}^{n} = \underbrace{p_{t}^{n} + a_{t}^{n}}_{=\delta_{t}^{n}} + X_{it}\gamma^{n} + U_{it}^{n} \tag{7}$$

$$w_{it}^{a} = \underbrace{p_{t}^{a} + a_{t}^{a}}_{=\delta_{t}^{a}} + X_{it}\gamma^{a} + U_{it}^{a} \tag{8}$$

In equations (7) and (8), p_t^j represents the price for each sector j at time t, and a_t^j stands for sectoral average productivity at time t. Together, they represent the returns from the sector-wide productivity at time t for sector j, denoted as δ_t^j. Hence, rewrite potential earnings for each sector as equations (9) and (10).

$$w_{it}^{n} = \delta_{t}^{n} + X_{it}\gamma^{n} + U_{it}^{n} \tag{9}$$

$$w_{it}^{a} = \delta_{t}^{a} + X_{it}\gamma^{a} + U_{it}^{a} \tag{10}$$

3.2 Absolute and Comparative Advantages

Comparative advantage is the difference in an individual's abilities between sectors. It is the differential returns based on this comparative advantage that affect sector decisions for economic agents. This comparative advantage comprises two components: observed characteristics (e.g. education), collected in the vector of X's, and unobserved traits–such as spatial intuition for arranging crops versus verbal dexterity in persuading clients–that differentially raise payoffs across sectors. Individuals know these latent abilities and internalize them when choosing sectors, while the econometrician does not as they are rarely recorded in data. As a result, ignoring this unobserved comparative advantage leads to selection bias in estimates of sectoral returns.

Therefore, I will introduce more structure to the unobserved error terms, U^j_{it}, in equations (9) and (10), reflecting the source of selection within the unobserved ability terms U^j_{it}. Following Lemieux (1998) and Suri (2011), decompose as in equation (11) and (12).

$$U^n_{it} = \theta^n_i + \xi^n_{it} \tag{11}$$

$$U^a_{it} = \theta^a_i + \xi^a_{it} \tag{12}$$

where θ^j_i is a time-invariant, sector-specific ability (permanent) and ξ^j_{it} is an idiosyncratic shock, which is unknown at choice, but follows a zero conditional mean (transitory). Hence, sorting depends on the permanent vector (θ^n_i, θ^a_i), not on ξ^j_{it}, shown in equation (13):

$$E(U^n_{it} - U^a_{it}) = \theta^n_i - \theta^a_i \tag{13}$$

Simply put, this transitory component of the error terms does not affect individuals' sector choices. Thus, when choosing a sector, individuals compare the difference of their expected potential earnings in each sector (w^n_{it} vs. w^a_{it}), which essentially compares the expected difference in the sector-wide productivity ($\delta^n_t - \delta^a_t$), rewards to the observed characteristics ($X's$), and the individuals' sector-specific latent abilities ($\theta^n_i - \theta^a_i$). To separate abilities that influence sector choice from those that do not, linearly project absolute advantage in each sector (θ^j_i) onto the "difference of latent abilities across sector" ($\theta^n_i - \theta^a_i$), shown as equations (14) and (15):

$$\theta^n_i = b_n(\theta^n_i - \theta^a_i) + \tau_i \tag{14}$$

$$\theta^a_i = b_a(\theta^n_i - \theta^a_i) + \tau_i \tag{15}$$

where τ_i is a common component (orthogonal to $\theta^n_i - \theta^a_i$) that moves productivity in both sectors equally (e.g., work ethic) and thus does *not* affect sector choice. The coefficients (b_n, b_a) are projection coefficients determined by the variance–covariance matrix of (θ^n_i, θ^a_i).[1]

[1]Closed-form expressions: $b_n = \frac{\sigma_n^2 - \sigma_{na}}{\sigma_n^2 + \sigma_a^2 - 2\sigma_{na}}$ and $b_a = \frac{\sigma_{na} - \sigma_a^2}{\sigma_n^2 + \sigma_a^2 - 2\sigma_{na}}$, where $\sigma_n = Var(\theta^n_i)$, $\sigma_a = Var(\theta^a_i)$, and $\sigma_{na} = COV(\theta^n_i, \theta^a_i)$.

Define the individual's *comparative advantage* as

$$\theta_i \equiv b_a(\theta_i^n - \theta_i^a), \tag{16}$$

and the *selection effect* as [2]

$$\beta \equiv \frac{b_n}{b_a} - 1. \tag{17}$$

Then individual sector-specific absolute advantages can be expressed as a function of comparative advantage and selection effect:

$$\theta_i^n = (1+\beta)\theta_i + \tau_i \tag{18}$$

$$\theta_i^a = \theta_i + \tau_i \tag{19}$$

Equations (18) and (19) are the key decompositions: (θ_i, β) describe the part of unobserved ability that *drives sorting* (comparative advantage and its sectoral loading), while τ_i is the sector-irrelevant component. This is precisely where TWFE regressions confound selection: they absorb *both* θ_i and τ_i into a single fixed effect, attributing common, sector-irrelevant traits to selection, such as hard work.

Hence, conditional on observables, sector choice is governed by the differential return to θ_i (the comparative advantage component), with loading $(1+\beta)$ in non-agriculture and 1 in agriculture. The common component τ_i is irrelevant for sorting.

In this formulation, the structural parameter β summarizes how strongly unobserved comparative advantage is rewarded *differentially* across sectors after netting out the correlation between sectoral abilities. Positive β indicates that the nonagricultural sector loads more on the relevant ability dispersion than agriculture; negative β indicates the opposite.

Intuitively, the parameter β measures how strongly differences in unobserved abilities are rewarded across sectors. When $\beta > 0$, variation in unobserved comparative advantage generates larger payoffs in non-agriculture than in agriculture. In this case, differences in individual talent translate more strongly into earnings outside agriculture, so workers with higher relative ability in non-agriculture have stronger incentives to sort into that sector, amplifying the observed productivity gap. When β is close to zero, the two sectors' value

[2] Later in this subsection, I'll further discuss why β is the selection effect.

differences in ability are similar, and sorting contributes little to the gap. Moreover, $\beta > 0$ means positive selection in the classical Roy model.[3]

Essentially, the selection effect and unobserved comparative advantages formulated by Lemieux (1998) capture the equivalent selection effect from unobserved heterogeneity that affects individuals' choices, as modelled in the classic Roy choice framework. The difference is that this formulation allows for the flexibility to estimate underlying unobserved comparative advantages avoiding distributional assumptions.

3.3 Main Empirical Model

After the discussion on β, let's return to the empirical model used in this paper. Building on equations (18)-(19), which decompose latent ability into comparative advantage and selection effect, I substitute into the potential earnings framework (equations (9) and (10)). I can rewrite the individual's log potential earnings at time t for each sector j in equations (20) and (21).

$$w_{it}^n = \delta_t^n + (1+\beta)\theta_i + \tau_i + X_{it}\gamma^n + \xi_{it}^n \tag{20}$$

$$w_{it}^a = \delta_t^a + \theta_i + \tau_i + X_{it}\gamma^a + \xi_{it}^a \tag{21}$$

Let D_{it} be a dummy variable, taking the value one if an individual i chooses the primary job in the nonagricultural sector at time t and zero otherwise. Then, I can write the individual's observed log earnings as equation (22).

$$w_{it} = D_{it}w_{it}^n + (1 - D_{it})w_{it}^a \tag{22}$$

where

$$D_{it} = \begin{cases} 1 & \text{non-agricultural sector} \\ 0 & \text{agricultural sector} \end{cases}$$

[3]The sign of β carries a selection interpretation: $\beta > 0$ corresponds to positive selection, $-1 < \beta < 0$ to negative selection, and $\beta < -1$ to a "refugee" case. A formal mapping of these regions to the selection types in the classic Roy model (Borjas, 1987)—together with the full algebra, the formal conditions, and a side-by-side comparison with Borjas—is developed in Chapter 3.

An individual chooses non-agriculture if she compares expected potential earnings in both sectors and finds that she would earn more in the non-agricultural sector, i.e., $w_{it}^n > w_{it}^a$; otherwise, she selects an agricultural job.

Then, substitute equations (20) and (21) in (22) to obtain equation (23).

$$\begin{aligned} w_{it} &= \delta_t^a + (\delta_t^n - \delta_t^a)D_{it} \\ &\quad + \theta_i + \beta\theta_i D_{it} + X_{it}\gamma^a + X_{it}(\gamma^n - \gamma^a)D_{it} + \tau_i + \epsilon_{it} \end{aligned} \tag{23}$$

where

$$\epsilon_{it} = D_{it}\xi_{it}^n + (1 - D_{it})\xi_{it}^a \tag{24}$$

Equation (23) is the main empirical model in this paper. I am interested in estimating the parameter β, the selection effect of comparative advantage, and recovering the distribution of comparative advantages θ_i. The fourth term in equation (23), $\beta\theta_i D_{it}$, contains the unobserved random variable θ_i, which is correlated with sectoral choice D_{it}; hence, this is a correlated random-coefficients (CRC) model. In this formulation, individual comparative advantage (θ_i) is explicitly defined as the individual's deviation from average sector productivity. If an individual works in the agricultural sector, sector-wide productivity is δ_t^a, and θ_i expresses how much more productive each individual is compared to the sector mean. If the non-agricultural sector is chosen, sector-wide productivity is represented by $\delta_t^a + (\delta_t^n - \delta_t^a)$, and the individual deviation from the sector mean is $(1 + \beta)\theta_i$.

This empirical framework models comparative advantage as the difference between absolute advantages across sectors, scaled by the covariance-adjusted spread, while distinguishing which unobserved abilities are relevant to sector choices. It directly addresses the two core challenges identified in the Literature Review—heterogeneity and endogeneity. Moreover, this formulation, consistent with the Roy model of sectoral choice, is better equipped to estimate individual sorting than a TWFE estimator or models that impose fixed effects, as reviewed previously, and allows for the estimation of selection effects without functional assumptions on latent abilities.

In a fixed-effects estimator using panel data, individual heterogeneity is absorbed at the person level rather than at the person-sector level, effectively imposing $\theta_i^n = \theta_i^a$ and thus $\beta = 0$. In other words, although the TWFE estimator corrects for time-invariant unobserved heterogeneity across individuals, it assumes that each individual's latent ability is identical across sectors and therefore cannot identify differential returns to unobserved comparative advantage. Moreover, by uniformly absorbing both the general ability component (τ_i)—which is irrelevant to sector choice—and the comparative advantage component (θ_i)—which is central to sorting—the fixed-effects estimator conflates the two and removes the very variation needed to identify β.

Structural approaches, by contrast, identify β by imposing a joint distribution on sector-specific abilities, e.g., (θ_i^a, θ_i^n) following a Normal or Fréchet distribution. In this setup, the selection effect arises from the assumed shape and correlation structure of that joint distribution: the dependence between sectors, and the relative spread of abilities, determines how much unobserved heterogeneity translates into differential payoffs across sectors. Chapter 3 illustrates this relationship formally by mapping the CRC selection term to the classical Roy model. Consequently, the estimate of β in this type of structural settings inherits these functional-form restrictions, rather than completely emerging from the observed variation in sectoral choices.

By contrast, following Lemieux (1998), the CRC model adapted in this paper relaxes those distributional assumptions by linearly projecting absolute advantage onto the latent abilities that determine sector choice (equations (18) and (19)). It retains the identity of the selection term β consistent with the classical Roy framework,[4] but achieves it through linear projection, thereby removing any functional-form requirement on θ_i. This relaxation makes it possible to use information contained in observed choice trajectories to infer unobserved comparative advantage θ_i while simultaneously identifying its impact, β.

Later in this chapter, I show how these choice trajectories reveal θ_i empirically and demonstrate how this framework recovers the distribution of comparative advantage without parametric assumptions. Before that, the next subsection discusses how the selection effect β contributes to the Agricultural Productivity Gap (APG).

[4]Chapter 3 provides the full mapping of β to the coefficient of the selection-correction term (the Inverse Mills Ratio) in a Roy model with a joint-normal distribution.

3.4 Individual Sorting and the APG

The objective of the main empirical model, as described in equation (23), is to estimate the structural parameter, the selection effect β, without imposing any distributional assumptions on latent skills. In the estimation section, I will describe how to obtain β without distributional assumptions. Now, suppose β is estimated from the data. In this subsection, I show that APG, as an observed object, can be estimated from the data under the perfectly competitive market assumption. Then, I demonstrate the composition of this observed APG. Finally, I show that the theoretical object of the selection term summarizes how much the selection effect contributes to the APG.

In this two-sector economy (agriculture and non-agriculture), assume that both are perfectly competitive. The output in each sector is given by equations (25) and (26).

$$Y_n = A_n H_n \tag{25}$$

$$Y_a = A_a H_a \tag{26}$$

where Y_j represents the sector aggregate output, A_j is the sector-specific efficiency, H_j is the efficient labour in each sector. Furthermore, the efficient labour H_j is the product of sector-specific human capital, h_j, and the total number of workers L_j in each sector j, represented by equations (27) and (28).

$$H_n = h_n L_n \tag{27}$$

$$H_a = h_a L_a \tag{28}$$

Following the agricultural productivity gap (APG) defined by Gollin et al. (2014), under the assumption of a perfectly competitive labour and goods market in both sectors, the wage in each sector equals the marginal value product of labour, and it also equals the average value of output per worker in each sector at equilibrium. Let W_j be the wage in sector j, and P_j be the price of good j. In equilibrium, APG–the ratio of value-added (VA) per worker between sectors j—can be expressed as the wage ratio at the sector level. $\frac{W_n}{W_a}$. See equation (29).

$$\underbrace{\frac{P_nY_n/L_n}{P_aY_a/L_a}}_{=\frac{VA_n/L_n}{VA_a/L_a}\equiv APG} = \frac{W_n}{W_a} \tag{29}$$

In the data, the difference in log earnings between sectors is APG, expressed in equation (30). This APG can be directly calculated by differencing the mean log earnings between two sectors, using the data.

$$log\left(\frac{W_n}{W_a}\right) = log(W_n) - log(W_a) = APG \tag{30}$$

Mathematically, take the expectation of the observed log wage w_{it} for each group, using the main empirical model expressed in equation (23), and then plug them into the APG equation (30). I can further rewrite APG as equation (31).[5]

$$APG = \Delta_\delta + \Delta_X + \Delta_\theta \tag{31}$$

Where the APG consists of three components: the sector-wide efficiency gap Δ_δ, as in (54); the observed average earnings disparity, Δ_X, given by (55); and the unobserved-ability contribution Δ_θ, as (56), between two sectors.

Furthermore, Δ_θ can be divided into three parts, as presented in equation (32): (i) the selection term, that is, the mean differential rewards for the non-agricultural workers when comparing what rewards they would have had if they had worked in the agricultural sector, given by S_θ; (ii) the group composition, which is, the gap in unobserved comparative advantage between the non-agricultural and agricultural sectors, denoted by d_θ; and (iii) the group difference in the latent abilities that are irrelevant to sectoral choices, d_τ.

$$\Delta_\theta = \underbrace{\beta E[\theta_i \mid D_i = 1]}_{\equiv S_\theta} + \underbrace{E[\theta_i \mid D_i = 1] - E[\theta_i \mid D_i = 0]}_{\equiv d_\theta} + d_\tau \tag{32}$$

Where d_θ captures the mean difference arising from unobserved comparative advantage—θ_i, given by (16)— between the non-agricultural and agricultural sectors. This term reflects differences in workers' unobserved comparative advantages. It completes the group profile on their earnings ability along with observed characteristics, such as education and gender.

[5] A step-by-step derivation of equation (31) is provided in Appendix 9.2.

In that sense, d_θ reflects the composition difference in unobserved characteristics between farmers and non-farmers. The mathematical expression of d_θ is shown in (33). This is not the effect of sorting, but it is an important object as the effect of observed characteristics would have been misunderstood if those latent abilities had not been accounted for.

$$d_\theta \equiv E[\theta_i \mid D_i = 1] - E[\theta_i \mid D_i = 0] \tag{33}$$

To answer the research question in this chapter, how much sorting into sectors explains the APG, the main interest is the term $\beta E[\theta_i \mid D_i = 1]$, denoted by S_θ in (34):

$$S_\theta \equiv \beta \, E[\theta_i \mid D_i = 1] \tag{34}$$

where β tells how much more or less differential reward there is between two potential earnings outcomes, that is, the selection effect, and $E[\theta_i \mid D_i = 1]$ captures the average unobserved comparative advantages for the non-agricultural workers. Hence, the share $\frac{S_\theta}{APG}$ captures the extent to which sorting contributes to the APG.

Moreover, d_θ is also interesting, as the group difference in unobserved comparative advantage completes the earnings profile beyond the characteristics that can only be observed in the data. Hence, the share $\frac{d_\theta}{APG}$ is a meaningful object, which measures how much the group composition difference in unobserved comparative advantage between non-agricultural workers and farmers contributes to the APG.

To recap, the main empirical model in this section explicitly models individual unobserved comparative advantage θ_i as a deviation from the sector-average productivity gap, thereby addressing unobserved heterogeneity. Moreover, it formulates the absolute and comparative advantages to allow for the estimation of the selection effect β without imposing a functional form on latent abilities.[6]

In what follows, I first aim to recover the structural parameters β and θ_i, and then aggregate them to assess the extent to which sorting contributes to the observed APG, $\frac{S_\theta}{APG}$, answering the main research question in this paper. When aggregating observed APG, I will also report how much the difference in group composition in their unobserved comparative

[6]The types of selection in this main model can be mapped to the selection types in the classic Roy model framework, which will be illustrated in Chapter 3.

advantage explains this gap, $\frac{d_\theta}{APG}$. Before getting into estimation strategies, I will first discuss the identification assumption in this CRC model.

4 Identification

The identification of this CRC model hinges on two key elements: First, the composite error terms $\tau_i + \epsilon_{it}$ is strictly exogenous. Second, the choice trajectory reveals unobserved comparative advantage θ_i.

4.1 A Strict Exogeneity Assumption

A strict exogeneity assumption of the composite error term, expressed in equation (35), delivers the identification for the main estimation equation (23).

$$E(\tau_i + \epsilon_{it}|\theta_i, D_{i1}...D_{iT}, X_{i1}...X_{iT}) = 0 \tag{35}$$

As τ_i represents individual i's unobserved abilities, regardless of sector choices, this strict exogeneity assumption is not overly restrictive for τ_i. Mathematically, equations (18) and (19) indicate that τ_i is removed from the comparative advantages, θ_i; hence, τ_i does not affect the sectoral choices and other observed regressors in (23).

The main concern arises from the transitory error term ϵ_{it}. While τ_i is time-invariant and unrelated to sectoral choice, ϵ_{it} could be correlated with D_{it} if short-term shocks (e.g., crop failure, weather events) influence both earnings and the decision to switch sectors. However, such shocks are typically transitory and do not justify abandoning a productivity-maximizing sector. Moreover, switching sectors often requires retraining or relocation, making it implausible as a short-run adjustment. Thus, individuals primarily adjust hours worked rather than change sectors in response to transitory shocks. As long as these temporary adjustments are controlled for, the strict exogeneity assumption remains credible.

However, some transitory shocks could impact individuals' realized earnings, such as crop failure due to extreme weather conditions, which could raise concerns about the identification of the model. This concern arises because individuals are likely to take measures to maintain their normal income level during adverse shocks, such as switching sectors or increasing the number of hours worked. Notably, individuals selected for their primary sector based on comparative advantage are unlikely to respond to short-lived, sector-specific shocks by changing sectors. This tendency to remain in the original sector is plausible for two reasons. First, the shocks are transitory and do not justify abandoning a productivity-maximizing allocation. Second, without retraining or skill upgrading, individuals are unlikely to achieve higher earnings in an alternative sector that lacks a comparative advantage.

If individuals choose to work more hours in response to temporary shocks, that could raise concerns about identification. This is because temporary shocks affect both working hours and individuals' earnings. To make the strict exogeneity assumption hold, such temporary shocks need to be controlled for. The next section will further examine if this is possible in the data.

Hence, in this setting, the exogenous shocks that cause individuals to switch sectors must be cost-related, such as making the pursuit of the alternative sector cheaper. For instance, as more factories are established in villages, the cost of securing a stable wage in the manufacturing sector is substantially reduced, and some people may be persuaded to switch sectors.

4.2 Revealed Comparative Advantage

If the strict exogeneity assumption on the composite error term holds, I can follow the projection approach by Chamberlain (1982, 1984) and Suri (2011) to disentangle the correlation between unobserved comparative advantage θ_i and choice variable D_{it} in the fourth term $\beta\theta_i D_{it}$ in the main estimation equation (23). This step allows a reduced-form regression in which the error terms have conditional mean zero and the coefficients are functions of the structural parameters. Then, using a minimum distance estimator (MDE), I can identify the selection effect β and recover the distribution of θ_i.

In Section 3.2, I have discussed that the comparative advantage formulation by Lemieux (1998) only relies on linear projection and relaxes the distributional assumption on θ_i. Here, I further discuss how to obtain unobserved θ_i, which is free of a specific distributional form. In this setting, individuals make decisions by comparing expected potential earnings between two sectors, and the unobserved comparative advantage θ_i contributes to the differential rewards in their earnings. Hence, θ_i is correlated with individual's choices D_{it}. Intuitively, individuals tend to stay in the sector in which their comparative advantages are stronger.

To separate θ_i and D_{it}, Chamberlain (1982, 1984) linearly project time-invariant θ_i on choice variable D_{it}, taking advantage of the fact that choice is made in each period. Further, Suri (2011) proposes adding the interactions to purge this correlation entirely. To operationalize the model, I follow Chamberlain (1982, 1984) and project the unobserved θ_i onto observed choice histories D_{it}, exploiting within-individual variation across periods. Equation (36) shows this step in a three-period panel.

$$\begin{aligned}\theta_i = {} & \lambda_0 + \lambda_1 D_{i1} + \lambda_2 D_{i2} + \lambda_3 D_{i3} + \lambda_4 D_{i1} D_{i2} \\ & + \lambda_5 D_{i1} D_{i3} + \lambda_6 D_{i2} D_{i3} + \lambda_7 D_{i1} D_{i2} D_{i3} + \nu_i \end{aligned} \tag{36}$$

As choice variable D_{it} is a dummy, this linear projection is a saturated model. Both Chamberlain and Suri call this step a purely technical procedure without any economic interpretation. This is because once I substitute Equation (36) in (23), the rearranged main estimation equation does not have θ_i, instead it is replaced by the function of choice variable D_{it} and their interaction terms. This substitution step allows two things: First, it allows a reduced-form regression, since each regressor is observed. Second, it links the coefficients of the reduced-form to the underlying structural parameters.[7]

Further on Equation (36), I propose an economic interpretation. Each period, individuals internalize their unobserved comparative advantages in order to decide on which sector is better rewarded. Hence, the choice histories reveal individuals' latent abilities. In the data, sector choices are observed, but comparative advantages θ_i are not. In my view, Equation (36) can also be a way to estimate unobserved comparative advantages by exploiting individuals' sectoral choice trajectories. So, I call the estimated unobserved comparative advantage, $\hat{\theta}_i$, the revealed comparative advantage. Appendix 9.4 further discusses this revealed comparative advantage concept and shows that choice trajectories contain such information in the data.

Together, strict exogeneity and the revealed comparative advantage substitution identify

[7]Appendix 9.3 illustrates this mapping in a simple two-period case, showing explicitly how reduced-form coefficients recover θ_i, mathematically.

the adapted CRC model. This approach allows estimation of the selection effect without assuming any specific distribution for latent abilities—unlike the Roy-type model, which hinges on joint normality. This revealed comparative advantage interpretation is what connects the structural framework to observable data, adding an economic meaning to a purely technical procedure.

Relative to the existing literature, this framework contributes by (i) redefining absolute advantage in line with the APG literature, (ii) introducing the notion of revealed comparative advantage to add economic interpretation to the projection method, and (iii) decomposing the APG into observed, productivity, and sorting components.

The next section applies this empirical model to data from the first three waves of the IFLS.

5 Estimate Selection on APG

The previous section described the empirical framework based on Suri (2011), extended to quantify the role of individual sorting in explaining sectoral productivity gaps. This section implements the model using the first three waves of the Indonesia Family Life Survey (IFLS), covering 1993–2000.[8]

I first describe the data and present descriptive statistics for the first three IFLS waves. Next, I estimate the reduced-form equations using Seemingly Unrelated Regressions (SUR) in the first stage, followed by recovering the structural parameters of interest—the selection effect β and the distribution of comparative advantage θ_i—in the second stage. Both stages are implemented using **randcoef**. Finally, I assess the contribution of individual sorting to the agricultural productivity gap (APG).

[8]The estimation uses the **randcoef** package in Stata, developed by Cabanillas, Michler, Michuda, and Tjernström (2018). The **tuple** package must be installed beforehand, both available via the Stata Community-Contributed programs repository.

5.1 Data

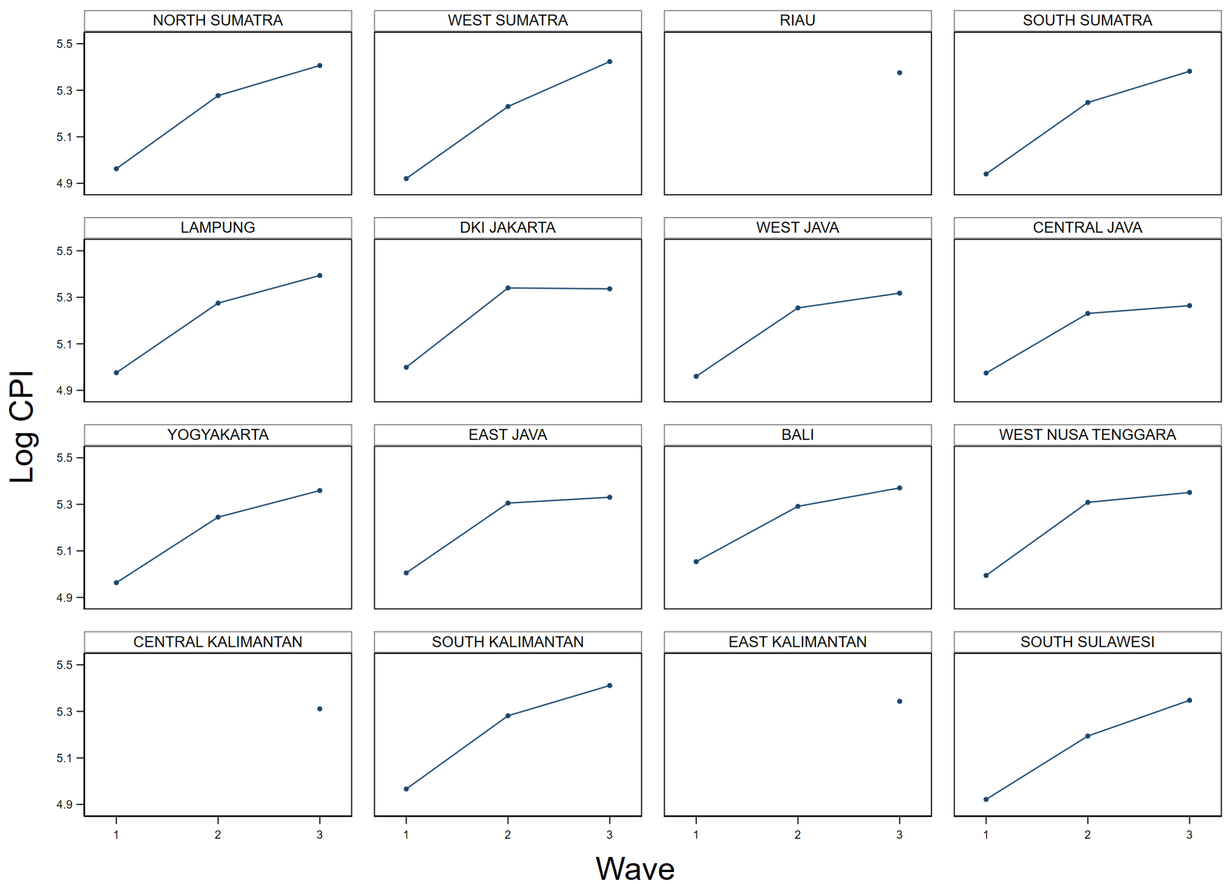
This study draws on two data sources: the Indonesian Family Life Survey (IFLS) and provincial consumer price indices (CPI) from Statistics Indonesia (Badan Pusat Statistik, BPS). The IFLS is a rich longitudinal household survey spanning five waves (1993, 1997, 2000, 2007, and 2014). The baseline covers 13 of the then–26 provinces, selected to represent roughly 83% of Indonesia's population and to capture broad socioeconomic and geographic diversity (Strauss et al., 2016). Over this period, Indonesia transitioned from a low-income to a lower-middle-income country. According to World Bank estimates (2025), real GDP per capita (constant 2015 USD) rose from $1,693 in 1993 to $3,171 in 2014, passing through the Asian Financial Crisis and entering a phase of sustained post-crisis recovery and growth. The IFLS provides detailed, repeated observations on individuals' employment, sectoral choices, earnings, and household characteristics, making it well suited to study how individual sorting contributes to sectoral productivity gaps.

During the IFLS period (1993–2014), Indonesia underwent major institutional and political changes. The authoritarian New Order regime collapsed with the ouster of President Suharto in May 1998, initiating the *Reformasi* transition to democratic governance (Freedom House, 1998). Subsequently, Parliament passed Laws 22/1999 and 25/1999, mandating a sweeping transfer of authority and revenues to regional governments, effective 2001—a shift known as the "Big Bang" decentralization (Hofman and Kaiser, 2002). Given the magnitude of this institutional break and its economic implications, this project naturally divides the IFLS data into two periods: 1993–2000 and 2000–2014. The current paper focuses on the first period, while a companion study will examine the later waves.

To account for spatial and temporal price variation, I incorporate province–year CPI data from Statistics Indonesia (BPS) as a control in all earnings regressions. Figure 1 plots the log CPI across the 16 provinces represented in the sample. The original IFLS baseline covered 13 provinces, but a small number of respondents moved to new provinces in later waves, producing limited observations for those areas (e.g., Riau, East Kalimantan, and Central Kalimantan). Overall, provincial price levels rose steadily between 1993 and 2000, with the sharpest increases outside Java — in West Sumatra, South Kalimantan and

North Sumatra — and the most muted in the Java core, notably Central Java, East Java and Jakarta. Because both earnings and CPI enter the regression in logarithmic form, the specification effectively adjusts for inflation and spatial price differences without explicitly deflating nominal earnings.

Figure 1: Log CPI



5.2 Descriptive Statistics

Because the empirical model requires observing both earnings and sector choice in every wave, I keep only individuals with complete information in all periods. The resulting panel contains 4,534 individuals followed over three waves, totalling 13,602 person-wave observations. These individuals come from 3,894 households in wave 1, 3,922 in wave 2, and 3,943 in wave 3; the small increase reflects household splits over time. About 27% of sampled individuals share a household with another sampled adult (i.e., multiple earners from the same household), and roughly 78% of the individuals are household heads.

Table 5 summarizes key characteristics (means/proportions) for the restricted and unrestricted samples. The restricted sample is systematically different, with more males (72% vs. 54%), slightly older respondents (mean age 44 vs. 40), and a higher marriage rate (90.8% vs. 75.9%).

This gender imbalance does not accurately reflect the gender composition of the full IFLS sample, but instead arises from sample restrictions: many women in the broader population work without pay in family enterprises or are out of the labour force as homemakers, and thus are excluded from the earnings-based analysis.

These restrictions arise mechanically from the need to observe both sectoral employment and earnings, rather than from survey attrition. In contrast, the sample remains broadly representative along key economic dimensions—education, average log earnings, average log hours worked, rural–urban composition, and the agricultural employment share are nearly identical to those in the unrestricted sample. While the restricted sample is not fully representative of the broader IFLS population—particularly along demographic dimensions—it is well suited for examining within-individual sectoral sorting, since it ensures consistent earnings and sector observations across all periods.

Tables 6 and 7 report summary statistics for the balanced analytical sample at the individual level. On average, individuals in the balanced panel were 40.3 years old in 1993, with mean age increasing steadily across waves as expected. Roughly 44% of the people resided in urban areas, a proportion that remained stable across survey rounds. This stability likely reflects the nature of the balanced sample, which includes only individuals with complete earnings and sector information in all three waves.

The share of individuals in non-agricultural work is relatively stable, declining slightly from 66.1% in 1993 to 63.7% in 2000. Approximately 46–41% report waged employment across waves,[9] implying a gradual rise in informality—from about 54% in 1993 to 59% in 2000. In this paper, sector is defined by the individual's primary job in each wave; both wage and self-employment (informal) are included within agriculture and non-agriculture.

[9] In this paper, I use 'wage' $\approx$ formal and 'non-wage' $\approx$ informal

Individuals reporting activities in both sectors are classified by the sector of their main job. Notably, non-agriculture has a much higher share of waged employment (54%) than agriculture (24%), and both sectors exhibit a decline in wage employment over time. Average monthly nominal income rose substantially—from IDR 125,698 in 1993 to IDR 449,344 in 2000—reflecting both real income growth and high inflation around the Asian financial crisis. This is corroborated by the rise in the Consumer Price Index (CPI), which climbed from 145.2 to 209.5 over the same period. Meanwhile, the average number of hours worked per month remained stable, hovering around 174 hours, with a slight decrease in working hours over time.

Table 8 reports household-level characteristics for the same balanced sample. Average household size increased modestly over time, from 4.78 members in 1993 to 5.79 in 2000. Family business ownership rose sharply—from 67% in the first wave to 80% by the third period. The share of households operating farm businesses remained relatively stable (rising from about 43% to 47%), while the share engaged in non-farm businesses increased significantly, from 38% to 52% over the three periods. Nominal asset values in both categories rose over time: farm business assets grew from IDR 7.0 million in 1993 to IDR 26.3 million in 2000, while non-farm business assets increased from IDR 6.4 million to IDR 9.2 million. Household assets not tied to any business—mainly real estate—were substantially higher, increasing from IDR 11.1 million to IDR 32.1 million. This comparison with property values suggests that most family businesses are relatively small in scale compared with overall household wealth holdings.

The IFLS also provides detailed information on economic shocks experienced by households in the five years preceding each survey. In 1993, 31.3% of households reported at least one shock, with an average of 0.39 shocks per household. In 1997, during the onset of the Asian financial crisis, 40% of households reported at least one shock, with an average of 0.56 shocks. By 2000, both the incidence (34%) and intensity (0.44 shocks on average) of reported shocks had declined.

The descriptive figures further illustrate income dynamics and sectoral choices in the dataset. Throughout the analysis, sector is defined by the individual's primary job in each wave; secondary activities are not used because the corresponding data are sparse and inconsistent. Figure 2 shows the distribution of log monthly income by wave and sector. Median log income rises over time in both sectors. In agriculture, the median increases from 10.5 in the first wave to 11.0 in the second wave and 11.9 in the third wave. In non-agriculture, it rises from 11.6 to 12.1 and then to 12.6 over the same waves. In each wave, the median is higher in non-agriculture than in agriculture, indicating a persistent sectoral income gap. The figure also shows that log earnings become more dispersed over the three waves, consistent with the rising standard deviation of log income reported in Table 6.

Figure 2: Log Income Non-ag vs. Agriculture over Three Waves (1993-1997-2000)

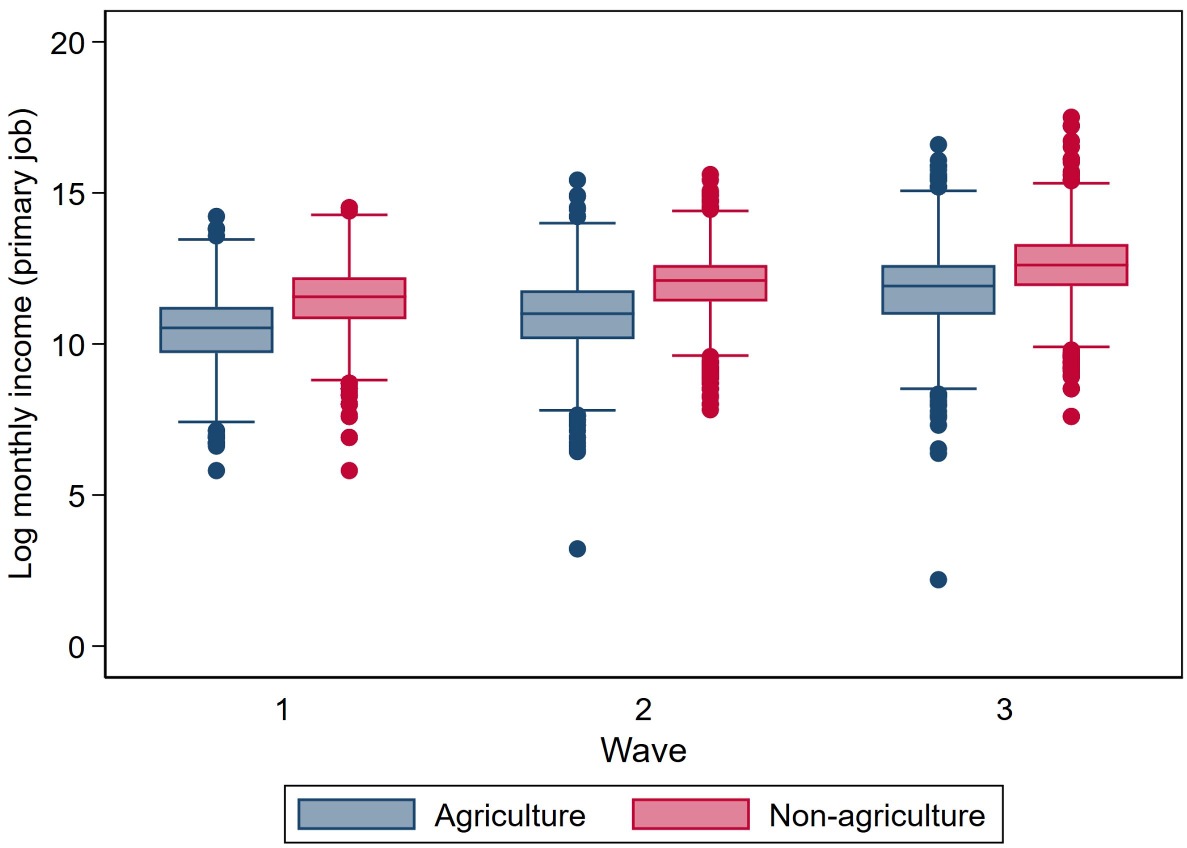


Figure 3 shows pooled average log earnings by sector and urban–rural location. Earnings are higher in urban areas for both sectors: average log earnings among non-agricultural workers are 12.19 in urban areas, compared with 11.75 in rural areas, while average log earnings among agricultural workers are 11.56 in urban areas and 11.00 in rural areas. Both locations exhibit a sizable sectoral earnings gap, but the gap is slightly larger in rural areas:

the earning difference between sectors is 0.75 log points in rural areas, compared with 0.63 log points in urban areas. Because location is closely related to both earnings levels and the sectoral earnings gap, I include an urban–rural indicator as a control in the reduced-form regressions.

Figure 3: Pooled Mean Log Earnings by Sector and Location

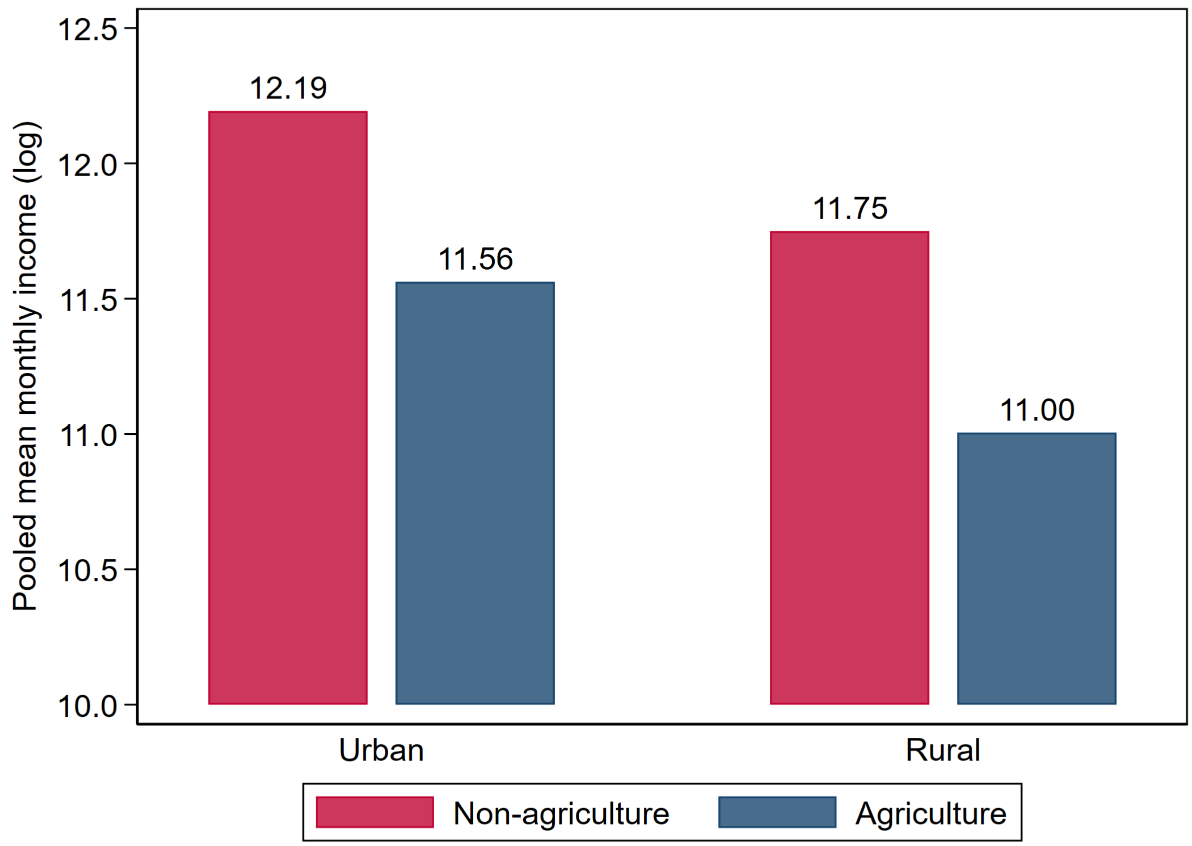
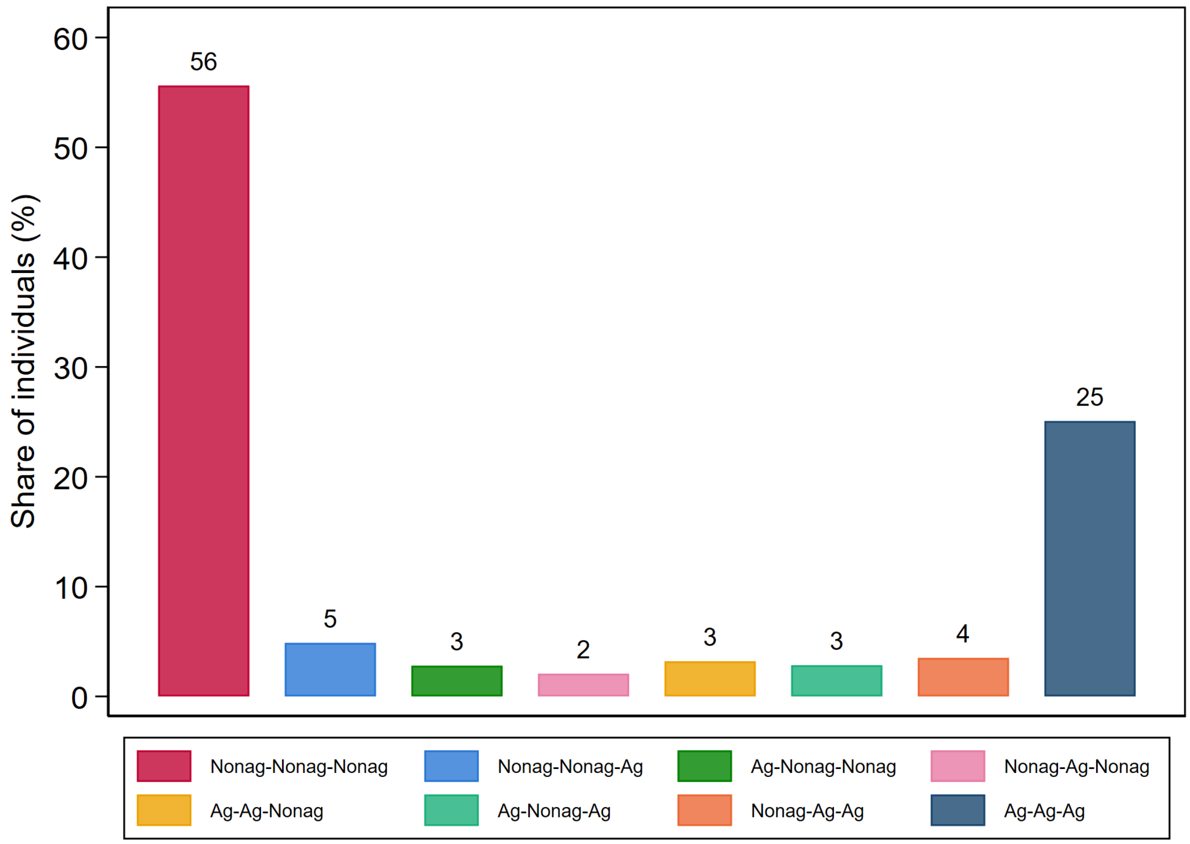


Figure 4 displays the distribution of sectoral transition patterns across the three survey rounds. A majority of individuals exhibit strong sectoral attachment: 56% remain in non-agriculture across all waves (Nonag–Nonag–Nonag), and 25% stay continuously in agriculture. In contrast, sector switchers constitute about 20% of the sample, with each group about 5% or less.

This persistence suggests that individuals make sectoral choices early in life—likely based on comparative advantage—and seldom alter them afterward. It lends empirical support to the assumption that unobserved sector-specific abilities, which drive sorting, are time-

Figure 4: Choice Trajectories over Three Waves



invariant over the study period. The stability of sectoral attachment may also reflect high costs of switching sectors, further discouraging mobility. At the same time, the roughly one-fifth of individuals who switch at least once provide meaningful variation essential for identifying the selection effect in the CRC model. The three-wave structure of the panel strengthens this identification by revealing not only whether individuals switch but also the direction and persistence of transitions (e.g., Ag–Nonag–Ag vs. Ag–Ag–Nonag), offering additional information to recover θ_i more precisely. Finally, when combined with the information on economic shocks across waves, these transition patterns reinforce the identification arguments developed in the previous section.

Complementing the evidence on sectoral persistence, Figure 5 presents the share of individuals engaged in non-agricultural employment by urban and rural location. Participation in non-agriculture consistently exceeds 85% in urban areas and remains around 45–50% in rural areas, with these spatial differences remarkably stable across survey rounds. Sectoral classification refers to individuals' main jobs only; while some workers hold secondary activities in the other sector, those are not included because secondary-job data are sparse and incomplete. Urban and rural designations follow the administrative classifications recorded in the IFLS (based on BPS definitions). The sizable share of non-agricultural employment in rural areas reflects the prevalence of non-farm opportunities outside cities and indicates that sectoral transitions need not always coincide with geographic migration. This challenges the common strategy of using rural–urban migration as a proxy for sectoral transitions in the Indonesian context.

Figure 5: Non-agriculture Share Urban vs. Rural over Three Waves (1993-1997-2000)

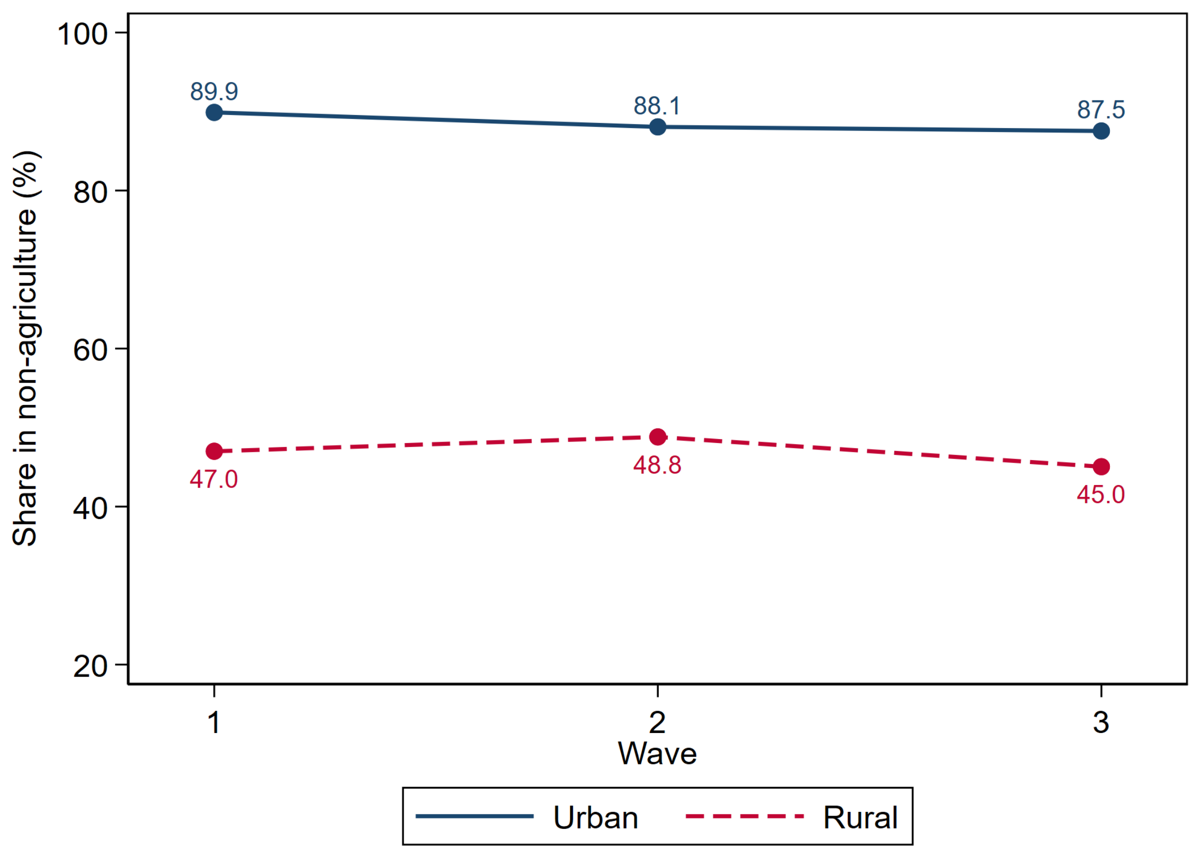
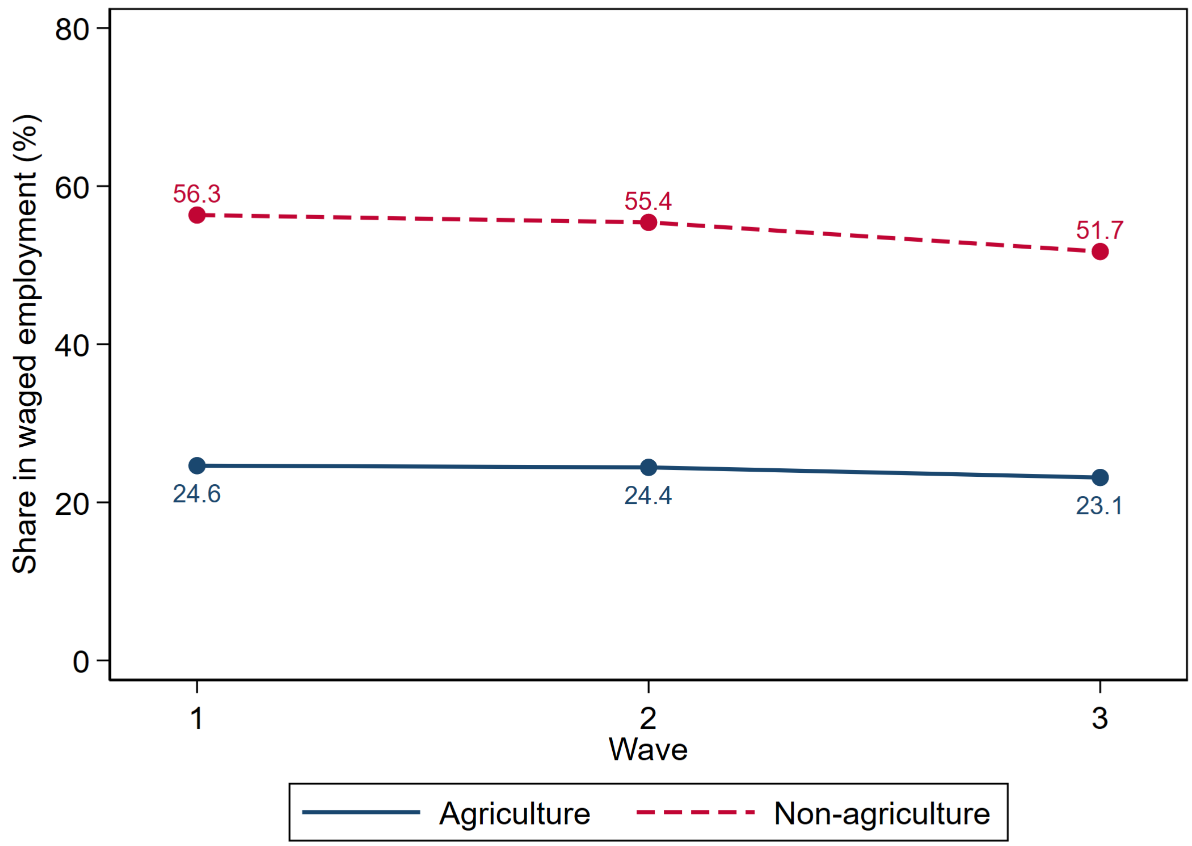


Figure 6 plots the share of wage-employed workers by sector. Waged employment accounts for just one-quarter of agricultural workers, compared with around half of non-agricultural workers. Both sectors show a mild decline in formal (waged) employment between 1993 and 2000, implying a corresponding rise in informality. This figure shows especially pronounced informality in agriculture, where roughly three-quarters of workers are self-employed.

Figure 6: Formal vs. Informal Workers (Wage employment ≈ formal; self-employed ≈ informal)



Together, these descriptive statistics highlight five key empirical patterns in the balanced panel from the first three IFLS waves: (1) income gaps between sectors are large and persistent, with distinct patterns across urban and rural areas; (2) earnings dispersion is substantial and widens over the period, and is consistently greater in agriculture than in non-agriculture; agriculture's long left tail primarily reflects genuine low returns to self-employment, with only a small residual of implausibly low values; (3) roughly 80% of workers remain in their initial sector—56% in non-agriculture and 25% in agriculture—while sector switchers account for about 20%; (4) sectoral composition differs sharply between urban and rural areas, a distinction explicitly controlled for in the regression analysis; and (5) informality rises in both sectors, and remains much higher in agriculture.

Together, these descriptive patterns also provide indirect evidence for the identification assumptions introduced earlier. In particular, the strong persistence of sectoral attachment and the stability of sector composition suggest that short-term shocks are unlikely to drive sectoral mobility. The next subsection examines this point directly by using IFLS data on households' exposure to and coping strategies for economic shocks.

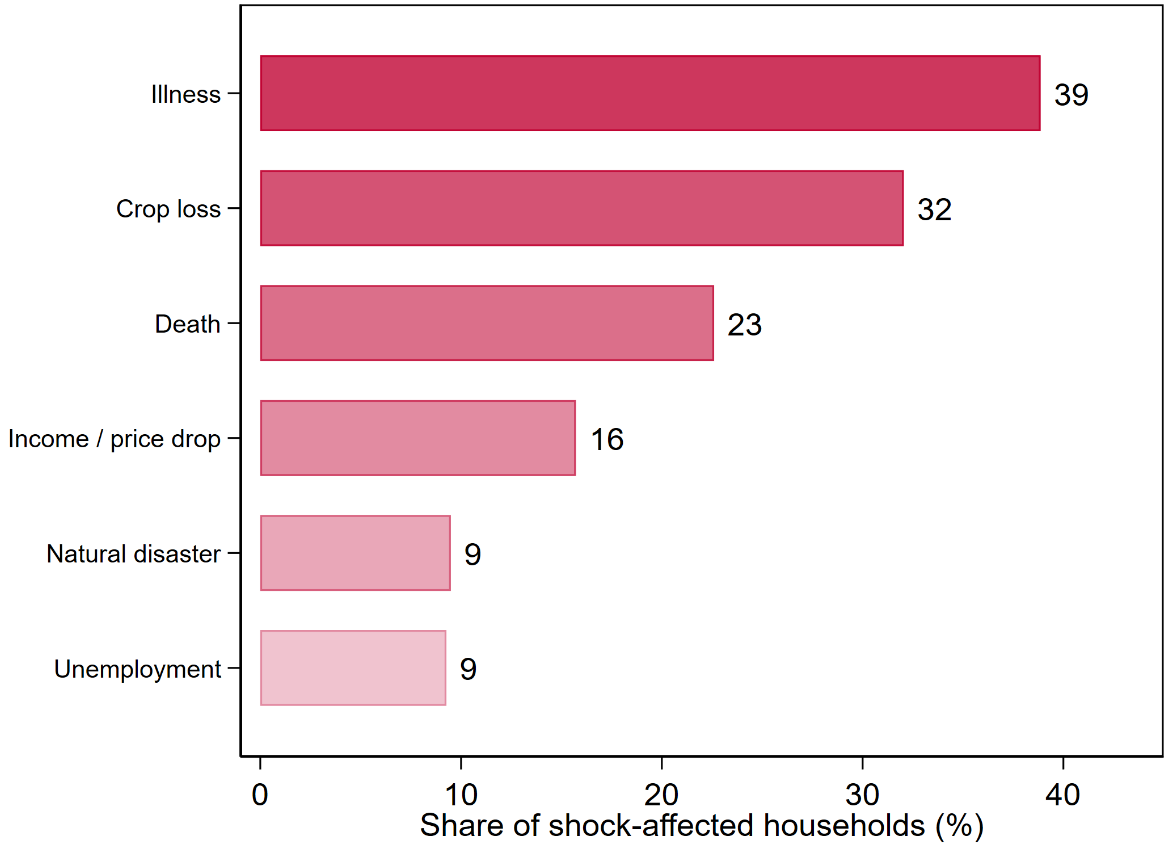
5.3 Empirical Evidence to the Identification Assumption

Empirical evidence from the Indonesia Family Life Survey (IFLS) supports the plausibility of the strict exogeneity assumption discussed in Section 4. In the first wave, respondents were asked whether they had experienced major economic shocks in the previous five years—such as crop failure, business loss, illness of a household member, or income loss (Figure 7)—and how they coped with those events (Figure 8).

The most frequently reported coping mechanisms were taking on additional jobs, borrowing or receiving transfers from relatives, using savings, and reducing expenditures. Notably, switching sectors does not appear among the recorded responses, indicating that individuals tend to remain in their primary sector and absorb short-term shocks through adjustments within the same occupation rather than by changing sectors. This pattern aligns closely with the behavioural mechanism underlying the identification strategy: sectoral choices are stable, and temporary shocks do not alter the long-run comparative advantage that drives selection.

Temporary shocks may still influence short-run earnings through changes in inputs such as hours worked or secondary activities. Since the IFLS provides information on major household-level shocks, these factors can be explicitly controlled for in the estimation. Including such variables mitigates potential correlation between shocks, labor supply adjustments, and earnings, thereby strengthening the credibility of the strict exogeneity assumption for the transitory error term ϵ_{it}. In practice, this ensures that observed income fluctuations associated with temporary shocks do not bias the estimation of the structural parameters of interest.

Figure 7: Types of Economic Shocks Reported by Households (IFLS Wave 1: 1993)

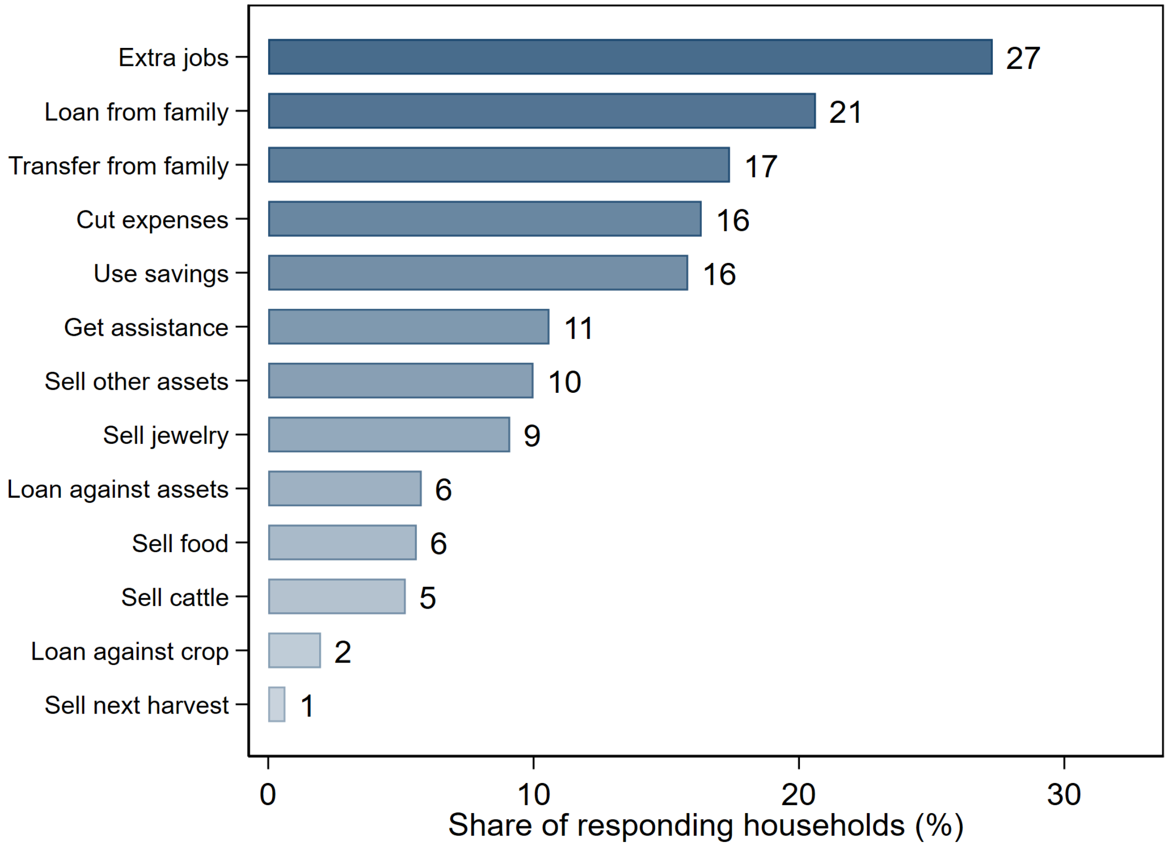


Together with the descriptive evidence presented earlier—showing limited sectoral mobility and high persistence in primary occupation—these findings lend empirical support to the identification assumptions of the model. The next subsection turns to estimating the reduced-form coefficients as the first step toward recovering the structural parameters.

5.4 Estimates Reduced-Form Coefficients

Section 3.3 introduced the main empirical model (23). Equation (36) projects the unobserved comparative advantage θ_i onto the choice histories. Substituting (36) into (23) yields

Figure 8: Coping Mechanisms Reported by Households (IFLS Wave 1: 1993)



reduced-form log earnings equation at each period.

$$\begin{aligned}
w_{it} = & \underbrace{\delta_t^a}_{\equiv\eta} + \underbrace{(\delta_t^n - \delta_t^a)}_{\equiv\alpha} D_{it} \\
& + \underbrace{\lambda_0 + \lambda_1 D_{i1} + \lambda_2 D_{i2} + \lambda_3 D_{i3} + \lambda_4 D_{i1} D_{i2} + \lambda_5 D_{i1} D_{i3} + \lambda_6 D_{i2} D_{i3} + \lambda_7 D_{i1} D_{i2} D_{i3} + \nu_i}_{=\theta_i} \\
& + \beta(\underbrace{\lambda_0 + \lambda_1 D_{i1} + \lambda_2 D_{i2} + \lambda_3 D_{i3} + \lambda_4 D_{i1} D_{i2} + \lambda_5 D_{i1} D_{i3} + \lambda_6 D_{i2} D_{i3} + \lambda_7 D_{i1} D_{i2} D_{i3}}_{=\theta_i \text{(continued below)}} \\
& \quad \underbrace{+\nu_i})D_{it} + X_{it}\gamma^a + X_{it}(\gamma^n - \gamma^a)D_{it} + \tau_i + \epsilon_{it}, \quad t = \{1, 2, 3\} \qquad (37)
\end{aligned}$$

To simplify calculation, let $\delta_t^a \equiv \eta$ and $\alpha \equiv \delta_t^n - \delta_t^a \; \forall t$. Equation (37) makes clear that the coefficients on the choice indicators and their interactions are functions of the underlying structural parameters. The observed characteristics X_{it} are included as covariates to improve precision. Rearranging terms yields the first-stage reduced-form regressions:[10]

$$
\begin{aligned}
w_{i1} &= \eta_1 + \phi_1 D_{i1} + \phi_2 D_{i2} + \phi_3 D_{i3} + \phi_4 D_{i1} D_{i2} + \phi_5 D_{i1} D_{i3} + \phi_6 D_{i2} D_{i3} + \phi_7 D_{i1} D_{i2} D_{i3} \\
&\quad + X_{i1}'\gamma_{11} + X_{i2}'\gamma_{12} + X_{i3}'\gamma_{13} + e_{i1}, && (38) \\
w_{i2} &= \eta_2 + \phi_8 D_{i1} + \phi_9 D_{i2} + \phi_{10} D_{i3} + \phi_{11} D_{i1} D_{i2} + \phi_{12} D_{i1} D_{i3} + \phi_{13} D_{i2} D_{i3} + \phi_{14} D_{i1} D_{i2} D_{i3} \\
&\quad + X_{i1}'\gamma_{21} + X_{i2}'\gamma_{22} + X_{i3}'\gamma_{23} + e_{i2}, && (39) \\
w_{i3} &= \eta_3 + \phi_{15} D_{i1} + \phi_{16} D_{i2} + \phi_{17} D_{i3} + \phi_{18} D_{i1} D_{i2} + \phi_{19} D_{i1} D_{i3} + \phi_{20} D_{i2} D_{i3} + \phi_{21} D_{i1} D_{i2} D_{i3} \\
&\quad + X_{i1}'\gamma_{31} + X_{i2}'\gamma_{32} + X_{i3}'\gamma_{33} + e_{i3}. && (40)
\end{aligned}
$$

Different from the ones in the Appendix 9.3 (which shows the two-period, no-covariate case), the reduced-form regression equations (38)-(40) add a third period, the corresponding interaction terms, and the X_{it} covariates as controls.[11]

The goal of the first-stage reduced-form regression is to estimate coefficients on the choice trajectories that map to the underlying structural parameters. To ensure that these estimates are not confounded, I include a comprehensive set of controls guided by the earlier descriptive analysis and identification assumptions. The covariates include indicators for urban/rural location, province, waged work (formal vs. informal), and household-level economic shocks, alongside standard observed characteristics such as age, education, marital status, hours worked, and gender. In addition, the logarithm of the provincial consumer price index (CPI) is included to account for spatial and temporal variation in price levels.

Tables 9, 10, and 11 report the reduced-form estimates for the 1993, 1997, and 2000 waves, respectively. Each table presents four specifications: Column (1) includes sectoral choices and their interactions across waves, without any controls. Column (2) adds location controls (urban/rural and province). Column (3) incorporates individual and job characteristics, including hours worked, waged work status, age, gender, marital status, religion,

[10]Appendix 9.3 presents the full algebraic derivation of the reduced-form regressions in a two-period case without covariates. Here, I simply state the reduced-form regressions in the first stage.

[11]The full set of interaction terms is written explicitly because the reported reduced-form tables (Tables 9–11) present the associated coefficients.

and education. Column (4) further adds province-level CPI (in logs) and a binary indicator for whether the household experienced any economic shock in the past five years. Because CPI is defined at the province level, province dummies are omitted in this full specification. Column (4) serves as the preferred model, while Column (1) provides a benchmark showing how coefficients shift when no controls are included.

Coefficients are in the first row for each regressor, and the corresponding standard errors are beneath them. A double-asterisk, **, denotes statistical significance at the 5% level, and a single-asterisk, *, indicates the 10% level. The end of each column reports the number of observations (N) and the R-squared value (R^2) for each regression. Five key patterns emerge from the reduced-form regressions:

1. Initial and contemporaneous sectoral choice as predictors of earnings: The results show a clear and stable pattern. The contemporaneous sector-choice indicator is significant in each period, indicating that earnings are strongly related to the sector in which an individual is currently employed. At the same time, the initial sector of employment remains significant across all three waves, suggesting that early sectoral placement captures persistent differences in earnings potential. In contrast, the interaction terms between initial and later sector choices are not statistically significant, indicating that switching histories do not add explanatory power beyond current sector and initial sector. This pattern is consistent with persistent sorting: earnings are shaped both by the current sectoral environment and by initial sectoral placement, while subsequent switching paths appear less important once these two components are controlled for. The result is consistent with the interpretation that time-invariant unobserved comparative advantage plays an important role in sectoral earnings differences.

2. Spatial differences in earnings are captured more consistently by the urban–rural distinction than by province fixed effects, although provincial location becomes relevant during the crisis-period wave. Urban residence is defined as a dummy variable equal to one if an individual resides in an urban area. Province is measured as a categorical variable covering 16 provinces: the 13 original IFLS provinces and 3 additional provinces observed because of household relocation. Cross-province relocation is rare

in the balanced sample, and fewer than 0.1% relocate to one of the three additional provinces. Across the specifications, urban residence remains an important predictor of earnings, while province generally adds less explanatory power once urban–rural location is accounted for. The exception is the crisis-period wave, when province becomes statistically significant, suggesting that broader regional conditions mattered more during that period. This pattern supports the preferred specification in Column (4), which treats urban–rural location as the main spatial source of earnings variation.

3. Urban-rural location in the initial period has significant explanatory power for earnings, but this explanatory power diminishes in the third period. This substantial impact of initial location is consistent with the low observed urban-rural mobility in the balanced sample, where the proportions are: 1,854 always-urban, 2,398 always-rural, and 281 switchers.

4. Hours worked exhibit a strong positive correlation with the earnings in the same period. Moreover, the dummy variable "wagedwork" captures the type of work for each individual, with a value of 1 for wage-paid jobs and 0 for self-employment. The type of employment has a significant impact on earnings, with formal employment paying higher (especially in 1993 and 1997), which aligns with the descriptive graph shown in the previous section.

5. CPI fluctuations align with economic shocks: The inclusion of log province-level CPI captures geographic price variation and some time-varying transitory shocks. After controlling for CPI, the economic shocks are not significant in the first period; however, they exhibit greater explanatory power for earnings in subsequent waves, suggesting that CPI may also absorb some of the transitory variation.

Finally, the covariates that commonly explain earnings behave as expected: age and education are consistently strong predictors of earnings. These results establish a credible reduced-form foundation for identifying structural parameters in the second-stage estimation. In addition, the point estimates are stable across the four specifications as the controls gradually increase.

Across specifications, R^2 rises as controls are added; standard earnings covariates (education, age, gender) enter with expected signs and magnitudes; and identification-motivated controls (urban/rural, CPI, shocks, hours, wage status) explain meaningful variation. Most importantly, the results show that sector-choice history matters, and that this history is captured mainly through two margins: initial non-agricultural status and contemporaneous sectoral choice. Their significant effects indicate that the first-stage reduced-form regression captures both persistent sorting from initial sectoral placement and earnings differences associated with current sectoral attachment. Taken together, these results give confidence that the coefficients on the choice indicators and their interactions contain the right information to recover the underlying structural objects (β, θ_i). I therefore map the reduced-form estimates onto the structural parameters and quantify the extent to which the APG is due to the selection effect β rather than sector-wide productivity α.

5.5 Recovering Structural Parameters

The second-stage estimation recovers the structural parameters that underlie the reduced-form coefficients—specifically, the selection effect (β) and the distribution of unobserved comparative advantage (θ_i).[12] [13]

As shown in the reduced-form equations (38)–(40), coefficients of choice at each period and their interaction terms ($\phi's$) are functions of the underlying structural parameters: sector-wide efficiency gap α, selection effect β, and λ_1–λ_7, where these λ's can be used to recover the distribution of unobserved comparative advantage θ_i. Recall the main estimation

[12] The estimation is implemented using the **randcoef** package in Stata (Cabanillas et al., 2018). The estimation in the second stage uses the minimum-distance estimator (MDE).

[13] The complete mathematical formulation of the second-stage calculation refers to the paper by Cabanillas, Michler, Michuda, and Tjernström (2018), who developed this **randcoef** STATA package.

equation with the substitution of the linear projection of θ_i:

$$
\begin{aligned}
w_{it} = &\underbrace{\delta_t^a}_{\equiv\eta} + \underbrace{(\delta_t^n - \delta_t^a)}_{\equiv\alpha} D_{it} \\
&+ \underbrace{\lambda_0 + \lambda_1 D_{i1} + \lambda_2 D_{i2} + \lambda_3 D_{i3} + \lambda_4 D_{i1} D_{i2} + \lambda_5 D_{i1} D_{i3} + \lambda_6 D_{i2} D_{i3} + \lambda_7 D_{i1} D_{i2} D_{i3} + \nu_i}_{=\theta_i} \\
&+ \beta(\underbrace{\lambda_0 + \lambda_1 D_{i1} + \lambda_2 D_{i2} + \lambda_3 D_{i3} + \lambda_4 D_{i1} D_{i2} + \lambda_5 D_{i1} D_{i3} + \lambda_6 D_{i2} D_{i3} + \lambda_7 D_{i1} D_{i2} D_{i3}}_{=\theta_i \text{(continued below)}} \\
&\quad \underbrace{+\nu_i})D_{it} + X_{it}\gamma^a + X_{it}(\gamma^n - \gamma^a)D_{it} + \tau_i + \epsilon_{it}, \quad t = \{1, 2, 3\} \qquad (37)
\end{aligned}
$$

Equation (37) displays all the structural parameters of interest: α, β, and λ_1–λ_7. Appendix 9.3 derives each coefficient's mathematical representation as the underlying structural parameters for a two-period case; extending to three periods simply adds one more period and additional interaction terms. Here, I only state the key structural parameters for clarity.

Table 1 presents the estimation results for these underlying structural parameters. Each column in Table 1 corresponds to one of the four model specifications used in the first-stage estimation (Tables 9–11). Column (1) excludes all covariates; Column (2) adds urban-rural residence and province; Column (3) further includes log hours worked, waged work, age, gender, education, religion and marital status; and Column (4) adds province-level log CPI and household-level economic shocks. The structural parameter estimates exhibit three key insights:

1. The selection effect (β) is small, negative, and *not* statistically significant. The point estimate of β captures the extent to which sectoral income differentials stem from individual sorting based on unobserved comparative advantages. Its values are negative in every specification, ranging from -0.115 in the baseline to -0.053 under the full controls.[14] Taken at face value, a negative β would indicate that the non-agricultural sector rewards unobserved comparative advantage *less* than agriculture does, so that individuals moving into non-agriculture are not those with the strongest latent non-agricultural advantage. This reading is consistent with the descriptive evidence that

[14] Reading across the columns of Table 1, β takes the values -0.115, -0.074, -0.152, and -0.053.

much of the movement into non-agriculture during this period reflects entry into lower-paid, informal work rather than positive sorting on non-agricultural ability. Moreover, the standard errors are large, ranging from 0.266 to 0.376, rendering the estimates statistically indistinguishable from zero. The data therefore do not support a meaningful aggregate selection effect.

Table 1: Structural Parameters

Structural Parameters	**(1)**	**(2)**	**(3)**	**(4)**
λ_1	0.261 **	0.171 **	0.171 **	0.174 **
	(0.059)	(0.058)	(0.054)	(0.053)
λ_2	0.002	-0.018	-0.101 *	-0.083
	(0.064)	(0.063)	(0.059)	(0.058)
λ_3	-0.117 *	-0.129 **	-0.105 *	-0.084
	(0.063)	(0.062)	(0.054)	(0.053)
λ_4	-0.095	-0.075	-0.032	-0.010
	(0.094)	(0.091)	(0.084)	(0.080)
λ_5	0.320 **	0.285 **	0.132	0.102
	(0.126)	(0.123)	(0.101)	(0.095)
λ_6	0.208 **	0.196 *	0.151	0.124
	(0.106)	(0.103)	(0.092)	(0.088)
λ_7	0.148	0.021	0.069	0.060
	(0.176)	(0.155)	(0.147)	(0.134)
α	0.370 **	0.387 **	0.336 **	0.350 **
	(0.101)	(0.086)	(0.071)	(0.064)
β	-0.115	-0.074	-0.152	-0.053
	(0.266)	(0.347)	(0.344)	(0.376)

Notes: Numbers in parentheses are standard errors. Asterisks indicate significance: * $p < 0.10$, ** $p < 0.05$.

This insignificant result aligns with the substantial variation in income observed in the data. As shown in Table 6, the pooled average monthly income from the primary job is IDR 260,461, while the standard deviation is IDR 694,561, about 2.7 times larger than the mean. Moreover, this pattern of widespread monthly earnings is persistent in each wave. This high dispersion in earnings likely implies high dispersion in latent abilities. Such dispersion means that even if selection is strong at the individual level for some, it may not translate into a substantial sectoral wage gap after considering the distribution of comparative advantages. Here is where the distribution assumption matters to the estimation results.

This finding highlights the critical role of the underlying distribution of comparative advantages. A more concentrated distribution could yield a more significant aggregate selection effect, whereas a widely dispersed distribution, as observed here, dilutes its statistical significance. Hence, distributional assumptions on unobserved comparative advantages raise concerns for the empirical estimation of the selection effect, even though it can offer profound theoretical insights.

2. The estimated distribution of comparative advantages (θ_i) deviates substantially from normality. Recall Equation (36) expresses the unobserved comparative advantages θ_i in a three-period model. After obtaining estimates of λ_1–λ_7, normalize θ_i such that $\sum \theta_i = 0$. Then, I can obtain λ_0 by calculating the intercept of θ_i, using Equation (41).

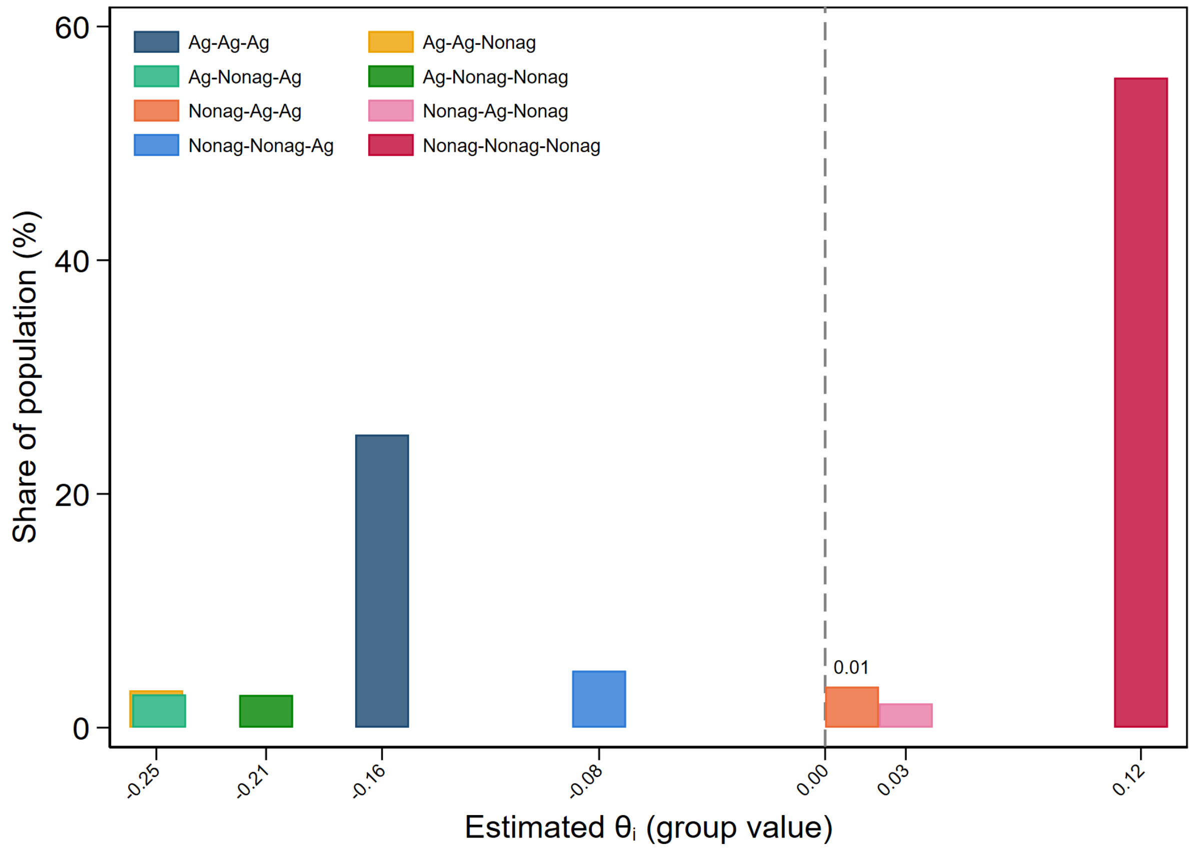
$$\begin{aligned} \theta_i &= \lambda_0 + \lambda_1 D_{i1} + \lambda_2 D_{i2} + \lambda_3 D_{i3} + \lambda_4 D_{i1} D_{i2} \\ &\quad + \lambda_5 D_{i1} D_{i3} + \lambda_6 D_{i2} D_{i3} + \lambda_7 D_{i1} D_{i2} D_{i3} + \nu_i \end{aligned} \tag{36}$$

$$\begin{aligned} \lambda_0 &= -\lambda_1 \overline{D_{i1}} - \lambda_2 \overline{D_{i2}} - \lambda_3 \overline{D_{i3}} \\ &\quad - \lambda_4 \overline{D_{i1} D_{i2}} - \lambda_5 \overline{D_{i1} D_{i3}} - \lambda_6 \overline{D_{i2} D_{i3}} - \lambda_7 \overline{D_{i1} D_{i2} D_{i3}} \end{aligned} \tag{41}$$

Once all the λ's are available, linear prediction recovers the empirical distribution of the revealed comparative advantages, $\hat{\theta}_i$. Figure 9 plots the recovered empirical distribution of $\hat{\theta}_i$ under specification (4) in Table 1. This figure has the normalized value of $\hat{\theta}_i$ labelled on the horizontal axis and the group share in the sample on the vertical axis. The distribution of comparative advantage is far from normal.

This finding on the empirical distribution of comparative advantage is consistent with a broader concern in the Roy-model literature: parametric assumptions about the joint distribution of latent earnings may be restrictive (Heckman and Honore, 1990). The recovered distribution is clearly bimodal, with the two peaks corresponding primarily to sectoral stayers: individuals always observed in agriculture cluster around the negative peak, while those always observed in non-agriculture cluster around the positive peak. Non-agricultural-origin switchers lie slightly to the left of non-agricultural stayers, consistent with weaker estimated comparative advantage in non-agriculture among those who later leave. Agriculture-origin switchers who move into non-agriculture, however, are concentrated at the most negative values of $\hat{\theta}_i$ below even agricultural stayers.

Figure 9: Recovered Distribution of Comparative Advantage for Specification (4) - with Full Controls



Taken at face value, this appears puzzling: some individuals with the strongest estimated latent advantage in agriculture are precisely those who leave the sector.

The resolution is that sectoral choice depends on total potential earnings, not on unobserved comparative advantage alone. Workers compare expected earnings across sectors, and those earnings reflect both observed characteristics and unobserved sector-specific ability. Table 12 shows that agriculture-origin switchers are younger and markedly better educated than agricultural stayers. A worker may therefore have strong latent ability in agriculture but still find non-agriculture more attractive if observed characteristics, especially education, are rewarded more strongly outside agriculture. For example, a farmer's child may have substantial farming-specific knowledge through family experience, local networks, and familiarity with agricultural production.

Yet if she also has more schooling than agricultural stayers, the return to that schooling may make non-agricultural work the higher-earning option. In this case, switching does not contradict the presence of agricultural comparative advantage; rather, it reflects the worker's comparison of total potential earnings across sectors.

3. Sectoral productivity differences (α) explain a substantial and statistically significant share of the observed earnings gap. The estimated α is conditional on the normalization $E[\theta_i] = 0$ (equivalently, $\sum_i \theta_i = 0$) imposed above. As shown in Table 1, the estimated α values across columns (1) to (4) are 0.370, 0.387, 0.336, and 0.350, with corresponding standard errors of 0.101, 0.086, 0.071, and 0.064; every estimate is significant at the 1% level. This consistent precision suggests that sector-wide productivity differences remain a robust, primary factor in explaining the APG, even after controlling for a rich set of covariates and selection. Importantly, as more controls are added, the associated standard errors contract, enhancing the precision and statistical significance of the estimates. Quantitatively, α is the sector-wide efficiency gap $\Delta_\delta \equiv \delta^n - \delta^a$ in the decomposition (31). Measured against an average observed APG of 0.953 log points (Table 2), the full specification implies that sector-wide efficiency accounts for $0.350/0.9528 \approx 36.7\%$ of the APG.

One might argue that the imprecision of the estimated β reflects weak inference arising from large variance or potential misspecification of the first-stage reduced form. However, the reduced-form regressions based on the Mincer framework exhibit reasonable explanatory power: standard covariates behave as expected, and the estimates remain stable as additional controls are introduced. Moreover, as shown in the Robustness section, the selection effect responds to winsorizing the data at the first and ninety-ninth percentiles. This sensitivity suggests that the estimate is shaped by the tails of the earnings distribution, as the CRC framework would predict, rather than being merely an artifact of a poorly specified first stage. The insignificance of β should therefore be interpreted as part of the model's flexible treatment of unobserved heterogeneity, rather than as simple evidence of noise alone.

Overall, the recovered structural parameters indicate that sector-wide productivity differences (α) are the dominant source of the observed APG. As the model incorporates richer controls, α becomes more precisely estimated, while the contribution of individual sorting (β) remains weak and empirically fragile. This fragility stems from the wide dispersion and non-normality of the recovered distribution of θ_i, which dilutes aggregate selection effects even when they may be present at the individual level.

5.6 Individual Selection on APG

Section 3.4 showed that the agricultural productivity gap in equation (31) can be decomposed additively into three components: the sector-wide efficiency gap Δ_δ, the contribution of observed characteristics Δ_X, and the contribution of unobserved factors Δ_θ. The preferred specification, Model (4) in the preceding section, provides the parameter estimates needed to construct the empirical counterparts of these components. This subsection combines those estimates to obtain the full APG decomposition and then examines the share of the APG attributable to selection on unobserved comparative advantage, $\frac{S_\theta}{APG}$.

Table 2 reports the empirical counterpart of each component in equation (31), constructed using the estimates from Model (4) in Section 5. For each wave, the APG is measured as the observed difference in mean log earnings between non-agricultural and agricultural workers. Under the normalization $E[\theta_i] = 0$, the sector-wide efficiency gap Δ_δ is represented by the estimated mean sector premium $\hat{\alpha}$ from Model (4) in Table 1. Because α is restricted to be common across waves, this component is constant over time.

The observed-characteristics component is constructed from the first-stage estimates reported in Tables 9–11. Equation (55) allows observed characteristics to have sector-specific returns, whereas the first-stage specification estimates a common coefficient vector $\hat{\gamma}$. Accordingly, I construct the empirical observed-characteristics component in wave t as follows:

$$\widehat{\Delta}_{X,t} = \hat{\gamma}' \left(\overline{X}_t^n - \overline{X}_t^a \right), \tag{42}$$

where $\overline{X}_t^n$ and $\overline{X}_t^a$ denote the mean observed characteristics of non-agricultural and agricultural workers, respectively. Thus, $\widehat{\Delta}_{X,t}$ measures the portion of the earnings gap associated with between-sector differences in observed characteristics, evaluated at the common returns estimated in the first stage.

The mean APG across the three waves is 0.952 log points. On average, observed characteristics account for 39.5% of the APG, while the sector-wide efficiency gap accounts for 36.7% under the normalization $E[\theta_i] = 0$. I recover the unobserved component residually as follows:

$$\widehat{\Delta}_{\theta,t} = \text{APG}_t - \widehat{\Delta}_\delta - \widehat{\Delta}_{X,t}. \tag{43}$$

The resulting unobserved component accounts for 23.8% of the APG on average. This component contains the selection term associated with unobserved comparative advantage. The analysis below therefore uses the structural estimates to construct S_θ and evaluate its magnitude relative to the APG.

Table 2: APG Decomposition

	$\mathbf{APG = \Delta_\delta + \Delta_X + \Delta_\theta}$			
	APG	$\boldsymbol{\Delta_\delta}$	$\boldsymbol{\Delta_X}$	$\boldsymbol{\Delta_\theta}$
Wave 1	1.056	0.350	0.377	0.330
Wave 2	1.020		0.395	0.275
Wave 3	0.781		0.356	0.076
Average	0.952		0.376	0.227
% of APG	**100.0%**	**36.7%**	**39.5%**	**23.8%**

Notes: Entries are in log points; the bottom row reports each component's share of the APG (%). Estimates from Model (4), $N = 4{,}425$.

By construction, the three components account fully for the observed APG. The sector-wide efficiency gap is quantitatively important, accounting for 36.7% of the APG under the maintained normalization. Observed characteristics constitute the largest component, accounting for 39.5% on average. Non-agricultural workers have, on average, more schooling and are more likely to reside in urban areas and engage in formal wage employment. Evaluated at the estimated common returns, these between-sector differences generate a substantial observed-characteristics component. Unobserved factors account for the remaining 23.8% of the APG.

Separating observed characteristics from latent abilities is important for interpreting Δ_X. Unobserved comparative advantage, θ_i, is an earnings-relevant trait that may be correlated with observed characteristics. If it is omitted, its contribution may be partly attributed to the estimated returns to observables, thereby confounding the observed- and unobserved-characteristics components. Distinguishing these components therefore requires estimating not only the contribution of observed characteristics but also the difference in mean unobserved comparative advantage between workers in the two sectors, denoted by d_θ.

As shown in equation (56), the unobserved component Δ_θ can be further divided into three parts: the selection term associated with sorting into the non-agricultural sector, S_θ; the between-sector composition difference in unobserved comparative advantage, d_θ; and the between-sector mean difference in latent abilities unrelated to sector choice, d_τ. Using the structural estimates from Model (4) in Table 1, I construct and report these components in Table 3 under the maintained normalization $E[\theta_i] = 0$.

The normalization has different implications for the three components. In particular,

$$S_\theta \equiv \beta E[\theta_i \mid D_i = 1] \tag{34}$$

depends on the level of $E[\theta_i \mid D_i = 1]$. Because this conditional mean cannot be recovered separately from the location normalization, S_θ is not point identified without such a normalization. By contrast, the composition difference

$$d_\theta = E[\theta_i \mid D_i = 1] - E[\theta_i \mid D_i = 0] \tag{33}$$

is invariant to a common location shift in θ_i, because the shift cancels when the two conditional means are differenced.

Under the maintained normalization, Table 3 shows that the selection term associated with unobserved comparative advantage is slightly negative, $\frac{S_\theta}{\text{APG}} = -0.47\%$. The mean difference in sector-neutral latent abilities is also negative: $\frac{d_\tau}{\text{APG}} = -0.76\%$.

Thus, both components marginally reduce, rather than enlarge, the APG. Together, they reduce the gap by approximately 1.2%. By contrast, the composition difference in unobserved comparative advantage, d_θ, accounts for 25.1% of the APG, as reported in the second column of Table 3. The three components therefore account for the total contribution of unobserved factors, $23.8\% \approx 25.1\% - 1.2\%$.

Importantly, d_θ is not itself the selection term. It measures the difference in mean unobserved comparative advantage between the actual non-agricultural and agricultural workers. By contrast, S_θ captures the earnings contribution arising from sectoral sorting based on the differential return to unobserved comparative advantage. Here, β measures the relative potential rewards to unobserved comparative advantage in non-agriculture. Thus, S_θ is the selection term associated with sorting on unobserved comparative advantage.

Table 3: Selection vs. Composition Effect

	$\boldsymbol{S_\theta}$ Selection	$\boldsymbol{d_\theta}$ Composition	$\boldsymbol{d_\tau}$ Sector-neutral	$\boldsymbol{\Delta_\theta}$ Total Unobserved
Wave 1	−0.005	0.276	0.058	0.330
Wave 2	−0.004	0.216	0.063	0.275
Wave 3	−0.004	0.224	−0.144	0.076
Average	−0.004	0.239	−0.007	0.227
% of APG	−0.47%	25.1%	−0.76%	23.8%

Notes: Entries are in log points; the bottom row reports each column's share of the APG (%). The three components satisfy $S_\theta + d_\theta + d_\tau = \Delta_\theta$; see Appendix 9.2 for the definition of each object. Estimates from Model (4), $N = 4{,}425$.

The separate identification of the sector-wide efficiency gap Δ_δ (equivalently, α) and the selection term S_θ depends on a location normalization for unobserved comparative advantage. Suppose the location of θ_i is shifted by an arbitrary constant c, so that

$$E[\tilde{\theta}_i] = E[\theta_i] + c \tag{44}$$

The sector-wide efficiency gap then changes by an offsetting amount: $\tilde{\alpha} - \alpha = -\beta c$. [15]

The selection term changes in the opposite direction. Because $S_\theta \equiv \beta E[\theta_i \mid D_i = 1]$, the

[15] To see the effect of this recentering on α, define $\tilde{\theta}_i = \theta_i + c$, so that $\theta_i = \tilde{\theta}_i - c$. Substituting this expression into equation (57), a stripped-down version of the observed wage equation in Appendix 9.3,

recentered selection term is as follows:

$$\begin{aligned}\widetilde{S}_\theta &= \beta E[\tilde{\theta}_i \mid D_i = 1] \\ &= \beta \left(E[\theta_i \mid D_i = 1] + c \right) \\ &= S_\theta + \beta c. \end{aligned} \tag{47}$$

Consequently,

$$\widetilde{\alpha} + \widetilde{S}_\theta = (\alpha - \beta c) + (S_\theta + \beta c) = \alpha + S_\theta. \tag{48}$$

Thus, although α and S_θ are individually identified only conditional on the location normalization, their sum is invariant to a common recentering of θ_i.

Hence, α is identified only conditional on the location normalization imposed on θ_i, and its sensitivity to a shift in the mean of the latent distribution is governed by

$$\Delta\alpha = -\beta c.$$

If $\beta = 0$, recentering the distribution of θ_i has no effect on α. More generally, the magnitude and direction of the change depend on both β and the location shift c. Model (4) reports $\hat{\beta} = -0.053$, with a standard error of 0.376. At the point estimate, a positive shift in $E[\theta_i]$ increases $\hat{\alpha}$ by $0.053c$, while a negative shift reduces it by the same proportional amount. Relative to an average APG of 0.952 log points, this implies that economically plausible shifts in the mean of unobserved comparative advantage produce only small changes in the estimated efficiency component. For instance, recentering θ by one standard deviation of its

gives

$$\begin{aligned} w_{it} &= \eta + \alpha D_{it} + (\tilde{\theta}_i - c) + \beta(\tilde{\theta}_i - c) D_{it} + u_{it} \\ &= \underbrace{(\eta - c)}_{\tilde{\eta}} + \underbrace{(\alpha - \beta c)}_{\tilde{\alpha}} D_{it} + \tilde{\theta}_i + \beta \tilde{\theta}_i D_{it} + u_{it}. \end{aligned} \tag{45}$$

The transformed parameters $(\tilde{\eta}, \tilde{\alpha}, \tilde{\theta}_i)$ reproduce exactly the same observed earnings as (η, α, θ_i) and therefore cannot be distinguished by the data. It follows that

$$\Delta\alpha \equiv \tilde{\alpha} - \alpha = -\beta c, \tag{46}$$

while β remains unchanged. The interaction term generates this result: substituting $\theta_i = \tilde{\theta}_i - c$ produces the additional term $-\beta c D_{it}$, which is absorbed into the coefficient on D_{it}.

recovered distribution ($\mathrm{SD}(\hat{\theta}) \approx 0.14$) changes $\hat{\alpha}$ by only $0.053 \times 0.14 \approx 0.008$ log points, less than 1% of the average APG of 0.952. I therefore report the APG decomposition in Tables 2 and 3 under the maintained normalization $E[\theta_i] = 0$, while interpreting the separate values of $\hat{\alpha}$ and $\widehat{S}_\theta$ conditional on that normalization.

In sum, the observed APG defined in Section 3.4 is accounted for by three components: the sector-wide efficiency gap Δ_δ, which accounts for 36.7%; observed characteristics Δ_X, which account for 39.5%; and unobserved factors Δ_θ, which account for the remaining 23.8%. Within Δ_θ, the between-sector composition difference in unobserved comparative advantage, d_θ, is the largest component:

$$\frac{d_\theta}{\text{APG}} = 25.1\%.$$

Under the maintained normalization $E[\theta_i] = 0$, the estimated selection term is slightly negative:

$$\frac{S_\theta}{\text{APG}} = -0.47\%.$$

This estimate indicates that the selection term makes a quantitatively small negative contribution under the maintained normalization.

6 Robustness

The previous section estimated the main empirical model, recovered the structural parameters, and decomposed the agricultural productivity gap. The main finding is that sector-wide productivity, α, accounts for a substantial share of the APG (36.7%), while the selection effect, β, is negative, economically modest, and statistically insignificant. Although the APG decomposition is reported only for the preferred specification with full controls, the key estimates are stable across the four baseline specifications, Models (1)–(4), reported in Section 5. This section examines whether these conclusions remain robust to alternative sample construction choices.

I proceed in three steps. First, I define three alternative samples and explain what each robustness check is designed to probe. Second, I review the first-stage reduced-form results and structural estimates, comparing them with the preferred specification from the previous section. Third, I compare the key parameters and APG decompositions across samples and draw out the implications of the robustness exercise.

6.1 Robustness Specifications

The preferred specification in Section 5 is Model (4), reported in Table 1 and Tables 9–11. In this robustness section, I refer to this preferred specification as "Main". It serves as the baseline against which the alternative data-treatment choices are compared.

In the main dataset, 81 earnings observations are reported as non-positive values and therefore have no valid logarithm. Further investigation shows that all 81 observations occur in the third wave and appear implausible when compared with the same individuals' earnings in other waves, hours worked, and job type. I therefore treat these observations as invalid and code their log earnings as missing in the main sample. Because the estimation requires a balanced panel, this treatment leaves 4,425 individuals in the main analysis. Each robustness specification changes one data-handling choice relative to this baseline. I consider three robustness specifications.

1. **Winsorized.** Starting from the main sample, log earnings in each wave are winsorized at the first and ninety-ninth percentiles. This caps the influence of extreme observations without dropping them, so the sample size remains unchanged at 4,425. Because the estimated selection effect depends on the dispersion of latent abilities, this specification probes the sensitivity of the results to the tails of the earnings distribution.

2. **Imputed.** This specification reconstructs the 81 invalid third-wave earnings observations by combining cross-sectional and longitudinal information. First, for each wave, I estimate a log-earnings prediction regression using the contemporaneous sector indicator, the full set of Model (4) covariates, and province dummies. The regression is fitted on the sample excluding the affected records, so that the invalid observations do not influence the prediction coefficients. This yields a predicted third-wave log earnings value for each affected individual based on their observed characteristics. Second, because individuals may differ persistently from what their observables predict, I add

a person-specific offset. This offset is defined as the average residual from the individual's first- and second-wave earnings prediction regressions, capturing the persistent component of earnings not explained by observed covariates. The imputed third-wave log earnings value is then the sum of the predicted value and the person-specific offset. This procedure uses both the earnings structure estimated from other individuals in the same wave and each individual's own earnings history in earlier waves. The resulting sample retains all 4,505 individuals for model estimation. I refer to this specification as "Imputed". It probes whether the main results are sensitive to treating the 81 third-wave earnings observations as missing.

3. **Assets.** This specification adds two household-wealth controls to the main sample: an indicator for whether the household operates a family business and the log of total household assets. I refer to this specification as "Assets". Because these characteristics are observed but excluded from the baseline specification, this robustness check examines whether part of the residual earnings variation attributed to unobserved ability is instead associated with observed household wealth. Requiring non-missing wealth information reduces the estimation sample to 4,302 individuals.

6.2 Robustness Estimations

Tables 13–15 report the first-stage reduced-form (SUR) estimates for the robustness specifications, and Table 16 reports the corresponding recovered structural parameters. In these tables, column (i) is the Main specification, while columns (ii)–(iv) report the Winsorized, Imputed, and Assets specifications, respectively.[16] Three features are worth noting.

1. The reduced-form estimates are generally stable in sign and magnitude across the four specifications. Initial sector choice remains an important predictor of earnings, and the contemporaneous sector choice is significant in every specification, though the third-wave effect is weaker throughout, generally attaining only the ten-percent level. The returns to hours worked and formal waged employment, as well as the

[16] To conserve space, the CPI and shock controls are folded into the indicator block in the reduced-form tables, and the province controls are omitted throughout, as in the main specification.

urban premium, largely reproduce the patterns observed in the main specification. The Assets specification produces somewhat larger changes in first-stage significance levels than the Winsorized or Imputed specifications, especially for urban residence and waged work in some waves. Even so, the general pattern of the reduced-form estimates remains consistent, and overall model fit is comparable across columns.

2. The structural estimates tell the same qualitative story across specifications. The sector-wide productivity gap, α, is positive, sizable, and statistically significant in every case: 0.350 in the Main specification, 0.324 in Winsorized, 0.363 in Imputed, and 0.359 in Assets. In contrast, the selection effect, β, remains negative and statistically insignificant throughout, with estimates of -0.053, -0.117, -0.050, and -0.012, respectively. The Winsorized specification produces the largest change in the magnitude of β, suggesting that the estimated selection effect is somewhat sensitive to the tails of the earnings distribution. However, this change does not alter the main conclusion: the estimated selection effect remains negative, economically modest, and statistically indistinguishable from zero.

3. The added wealth controls behave largely as expected. Log household assets enter strongly and significantly, whereas the family-business indicator is small and statistically insignificant, consistent with most family businesses being small-scale activities. Including these controls modestly improves model fit and changes some first-stage significance levels, but it does not alter the choice-history structure used to recover the structural parameters. The stability of α and β in the Assets specification suggests that observed household wealth does not account for the main structural findings.

6.3 Key Parameters and the APG Decomposition

Table 4 summarizes the key structural parameters and the APG decomposition across the four specifications. The comparison reinforces the main conclusion from the preferred model: the sector-wide productivity component remains a substantial and statistically significant contributor to the APG, while the selection effect remains small, negative, and statistically insignificant. I draw out three patterns from this comparison.

The first pattern is the significance and stability of the sector-wide productivity component. Expressed as a share of the observed APG, this component accounts for roughly 35 to 38 percent of the gap across all specifications. The observed APG itself is also stable, ranging only from 0.934 to 0.957 log points. Thus, neither winsorizing the earnings distribution, imputing the affected third-wave earnings observations, nor adding household-wealth controls changes the central finding that sector-wide productivity differences explain a substantial share of the agricultural productivity gap.

The second pattern concerns the selection parameter. The estimate of β remains negative in all four specifications and is never statistically significant. As noted above, its magnitude is close to zero in the Main, Imputed, and Assets specifications; however, the Winsorized specification produces a larger negative estimate, -0.117, and a slightly smaller standard error. This suggests that the estimated selection effect is somewhat sensitive to the tails of the earnings distribution, which is expected given that the selection parameter depends on the dispersion of latent abilities. However, the substantive conclusion is unchanged: there is no evidence of a large positive selection effect.

The third pattern appears in the Assets specification. Adding household wealth controls shifts part of the APG from the unobserved to the observed component. The share explained by observed characteristics, Δ_X, rises from 39.5 percent in the Main specification to 47.8 percent in the Assets specification. At the same time, the unobserved component, Δ_θ, falls from 23.8 to 14.6 percent, and the corresponding composition term, d_θ, falls from 25.1 to 15.4 percent. This indicates that part of what appears as unobserved heterogeneity in the baseline model is associated with observed household wealth once these controls are added. Importantly, however, controlling for wealth does not meaningfully change the estimated selection parameter: β moves closer to zero and remains statistically insignificant.

Table 4: Robustness: Key Parameters and APG Decomposition

	Main		**Winsorized**		**Imputed**		**Assets**	
	Level	%	Level	%	Level	%	Level	%
β	-0.053		-0.117		-0.050		-0.012	
	(0.376)		(0.343)		(0.397)		(0.466)	
$\Delta_\delta = \alpha$	0.350^{**}	36.7	0.324^{**}	34.7	0.363^{**}	37.9	0.359^{**}	37.6
	(0.064)		(0.065)		(0.062)		(0.065)	
Δ_X	0.376	39.5	0.371	39.8	0.378	39.5	0.456	47.8
Δ_θ	0.227	23.8	0.239	25.6	0.216	22.6	0.139	14.6
APG	0.952		0.934		0.957		0.953	
d_θ	0.239	25.1	0.255	27.3	0.227	23.7	0.147	15.4
N	4,425		4,425		4,505		4,302	

Notes: Columns correspond to specifications (i)–(iv) in the Robustness first-stage and structural tables: (i) Main; (ii) Winsorized; (iii) Imputed; (iv) Assets. Entries are in log points; standard errors in parentheses. ** $p < 0.05$, * $p < 0.10$.

Despite the informative role of the Assets specification, I do not use it as the preferred model for two reasons. First, the wealth variables are measured at the household level, while the earnings model is estimated at the individual level. As discussed in the data section, about 27 percent of individuals in the sample live in shared households, so household assets may not map cleanly onto each individual's own economic position. Including household wealth as an individual-level control therefore risks assigning the same household-level resource measure to multiple individuals whose labour-market choices and earnings may differ. Second, adding the wealth controls further reduces the balanced estimation sample because of missing asset information. For these reasons, I treat the Assets specification as a robustness check rather than the baseline specification: it is useful for assessing whether observed household wealth absorbs part of the residual heterogeneity, but it is less suitable as the main specification for the paper.

Taken together, the robustness checks leave the main conclusions intact while refining their interpretation. Sector-wide productivity differences consistently explain a substantial share of the APG, while the selection effect remains weak and imprecisely estimated across alternative data treatments. The Assets specification shows that some of the baseline unobserved component reflects observed household wealth omitted from the preferred specification, and the Winsorized specification shows that the selection estimate responds to the tails of the earnings distribution. These patterns motivate the discussion that follows, which considers how distributional features of earnings and latent ability shape the interpretation of selection in this setting.

7 Discussion

Contrary to the prevailing consensus in the APG literature, which attributes a substantial share of sectoral productivity gaps to self-selection, this chapter finds that individual sorting contributes minimally to the APG. Moreover, this study identifies a persistent and substantial sector-wide efficiency difference as a crucial driver of the sectoral earnings gap. Hence, the combination of sector-wide technology differences, individual observed heterogeneity, and unobserved comparative advantages explains the majority of sectoral productivity gaps when using a balanced panel in the first three waves of the IFLS data.

This conclusion departs from a body of literature that reports sizable selection effects. For example, Pulido and Świecki (2019) attribute 45–70% of the APG to sorting on the IFLS data under a joint-normality assumption. Using the same five-wave IFLS dataset, Hamory et al. (2021) estimate a ∼67% selection effect on APG by applying a fixed-effect approach. I argue that this divergence is largely methodological. The two approaches that dominate the literature—two-way fixed effect (TWFE) and parametric selection corrections in the spirit of the Roy (1951) model—each, by construction, tend to overstate the share of the gap attributable to selection. Understanding why clarifies how the present estimates relate to earlier findings.

The first approach, TWFE, is problematic when its sector coefficient is interpreted as selection on comparative advantage, for two reasons.

First, the identifying variation comes only from switchers. For the roughly 80 percent of workers who never change sectors, the sector-choice indicator D_{it} is time-invariant and is absorbed by the individual fixed effect. The estimated sector coefficient is therefore identified by the roughly 20 percent of workers who switch sectors. These switchers may differ systematically from stayers who are relevant for measuring the APG. As a result, using within-person earnings changes among switchers to infer the aggregate role of selection in the APG is questionable.

Second, TWFE absorbs all time-invariant individual heterogeneity into a single fixed effect. In the main model, permanent unobserved ability decomposes into a sector-relevant component, θ_i, which captures unobserved comparative advantage, and a latent sector-irrelevant ability, τ_i, which affects earnings similarly across sectors. A fixed effect absorbs both components without distinguishing unobserved comparative advantage from general individual traits such as work ethic, family background, or networks that raise earnings in either sector. This matters because only the sector-relevant component reflects selection on unobserved comparative advantage; the sector-irrelevant component reflects unobserved heterogeneity, but not selection in the Roy sense.

More broadly, a mean difference in unobserved time-invariant earnings components across sectors should not itself be interpreted as the selection effect. This distinction is visible in the decomposition results. Section 6 shows that, in the main specification, the composition term d_θ, which captures the group mean difference in unobserved comparative advantage, accounts for 25.1 percent of the APG. When household wealth controls are added in the Assets robustness specification, this share falls to 15.4 percent. This decline shows that part of what appears as unobserved heterogeneity in the baseline model is associated with observed household wealth omitted from the preferred specification. A fixed-effect approach would absorb this variation into a single individual effect, but it would not reveal whether the variation reflects sector-relevant comparative advantage, sector-irrelevant ability, observed wealth, or other persistent characteristics.

The CRC approach addresses this limitation by modelling sector-specific latent abilities directly. Instead of absorbing all permanent individual traits into one undifferentiated fixed effect, it models unobserved comparative advantage as the component of latent ability that is relevant for sector choice and potential earnings differences across sectors. The framework therefore separates sector-choice-relevant unobserved ability from latent traits that affect earnings similarly in both sectors. This is why the CRC selection parameter, β, can be interpreted as a selection effect, whereas the TWFE coefficient cannot.

The second approach recovers a closed-form selection parameter, but only after imposing a distributional structure—most commonly joint normality—on latent abilities across sectors. The joint-normal assumption on (θ_i^a, θ_i^n) makes the selection terms analytically tractable, but it also means that the shape of the latent-ability distribution is imposed rather than recovered from the data. In contrast, the CRC framework used here does not require a parametric distribution for unobserved comparative advantage. Instead, it infers the empirical distribution of θ_i from individuals' sectoral choice trajectories from both stayers and switchers.

As shown in Figure 9, the recovered distribution deviates substantially from normality, with mass concentrated in both tails. Although this approach relies on a discrete support of estimated coefficients (λ_1–λ_7) rather than a continuous density, it allows the data to reveal the heterogeneity pattern empirically instead of forcing it to conform to a pre-specified distribution. The Winsorized robustness specification in Section 6 reinforces this point. When the tails of the earnings distribution are compressed, the estimated selection parameter becomes more negative, moving from -0.053 to -0.117, although it remains statistically insignificant. This sensitivity is informative: it shows that the selection estimate responds to the distributional shape of earnings and latent ability. The small and imprecise aggregate selection effect is therefore not simply a consequence of one specification choice, but reflects the empirical distribution of comparative advantage recovered by the CRC model.

Taken together, these two points help reconcile my results with the literature. Larger selection effects reported elsewhere are consistent with methods that either fold sector-irrelevant ability into selection, as in TWFE interpretations, or impose distributional structure that can amplify selection effects, as in parametric correction approaches. By contrast, an approach that separates θ_i from τ_i and lets the data discipline the distribution of comparative advantage finds selection to be minor in 1990s Indonesia. The disagreement is therefore partly about method and identifying assumptions, not simply about the underlying labour market.

This comparison also clarifies why the CRC framework is well suited to the chapter's research question. By explicitly modelling unobserved comparative advantage, the CRC framework distinguishes sector-relevant comparative advantage from sector-neutral ability, so that the object it recovers corresponds conceptually to selection on unobserved comparative advantage in the literature rather than to an undifferentiated individual fixed effect. It draws on information from both stayers and switchers, avoiding reliance on a small switcher subsample. It treats sectoral choice trajectories as revealed signals of comparative advantage, giving the estimation a transparent economic interpretation. Finally, it recovers the distribution of latent ability without imposing a parametric prior, so that the estimated selection effect is disciplined by the data rather than by functional form.

These advantages come with genuine limitations. Identification relies on variation in choice trajectories and earnings over time: as Tjernström et al. (2023) emphasize, when earnings are similar across choice groups or when transitions are sparse, the system of moment equations becomes weakly identified and can break down. Moreover, this method requires a balanced panel and assumes that the relevant component of unobserved comparative advantage is time-invariant over the estimation period, which restricts the analysis to individuals observed in all three waves and rules out learning or life-cycle changes in individuals' comparative advantage. Finally, the approach identifies the relative reward to unobserved comparative advantage only up to a normalization, so the level of the selection payoff is not separately pinned down in the same way as the sector-wide productivity and composition components. For these reasons, the estimates are best read as evidence for Indonesia's pre-decentralization decade, under the maintained assumption of stable comparative advantage, rather than as a universal claim.

In the IFLS setting, the evidence points to sector-wide productivity differences rather than selection on latent skills as the primary driver of the APG. The policy implication is not that sectoral mobility is irrelevant, but that reallocation alone is unlikely to close the gap if agricultural productivity fundamentals remain weak. The relevant margins are therefore likely to include technology adoption, input and output market access, infrastructure, and management or extension services. I return to these implications in the conclusion, where I discuss policy margins consistent with the Indonesian context and the empirical patterns documented above.

8 Conclusion

This paper revisits a core question in the agricultural productivity gap literature: how much of the APG can be explained by individual sorting on unobserved comparative advantage? Using the first three waves of the Indonesia Family Life Survey, I focus on Indonesia's pre-decentralization decade, before the institutional changes associated with the 2000 "Big Bang" reforms. The evidence shows that sorting on unobserved comparative advantage plays only a limited role in explaining the sectoral earnings gap during this period. The estimated selection parameter is small, negative, and statistically insignificant. In the APG decomposition, sector-wide productivity differences and observed worker characteristics account for most of the gap, while the average difference in unobserved comparative advantage across sectors should not be interpreted as the selection effect itself.

This finding differs from much of the existing APG literature, but the difference is partly methodological. The CRC framework used in this paper separates sector-relevant unobserved comparative advantage from sector-neutral latent ability, uses information from both stayers and switchers, and avoids imposing a parametric distribution on latent abilities. These features matter because prevailing approaches can misattribute part of the gap to selection. TWFE absorbs all time-invariant heterogeneity into a single individual effect, and its interpretation often reads the reduction in the sector coefficient after adding fixed effects as evidence of selection. But this reduction reflects an undifferentiated bundle of persistent traits, not necessarily selection on unobserved comparative advantage as defined in the Roy framework. By recovering the empirical distribution of comparative advantage from observed choice histories, the CRC framework shows that larger unobserved heterogeneity does not necessarily imply a large selection effect.

The results also carry a clear policy implication. If selection on unobserved comparative advantage explains little of the APG, then reallocating workers out of agriculture is unlikely, by itself, to close the sectoral earnings gap. This does not mean that mobility is irrelevant; rather, it means that the relevant constraint is not only who works in each sector, but also the productivity environment in which they work. For Indonesia in the 1990s, the evidence points toward sector-wide productivity differences as a central margin. Policies that raise productivity, especially within agriculture, through technology adoption, input and output market access, infrastructure, and extension services are therefore more consistent with the empirical patterns documented here than policies premised mainly on sorting workers into non-agriculture.

More broadly, the chapter qualifies the Lewis-style interpretation of structural transformation as a simple movement of labour out of low-productivity agriculture. In this setting, the APG does not appear to persist mainly because low-ability workers remain in agriculture or because high-ability workers sort into non-agriculture. Instead, it reflects a combination of sector-wide productivity differences, observed worker characteristics, and unobserved composition differences, with little evidence that selection on unobserved comparative advantage widens the gap. The central lesson is that how the APG closes matters: convergence through productivity growth within sectors has different implications from convergence through labour absorption into low-productivity non-agricultural work. Extending the analysis to Indonesia's post-decentralization period would show whether these margins changed as the economy continued to transform.

Table 5: Comparison Between Restricted and Unrestricted Samples (IFLS Waves 1–3)

Variable	**Restricted Sample**	**Unrestricted Sample**
Age (years)	44	40
Male (%)	72.4	53.7
Married (%)	90.8	75.9
Education level (%)		
Unschooled	14.1	15.7
Primary	51.7	45.8
Junior high	11.2	13.1
Senior high	15.8	17.2
College / university	6.2	6.2
Others	1.0	2.0
Urban (%)	44.2	46.2
Non-agriculture (%)	65.3	64.0
Waged work (%)	44.0	42.6
Log earnings	11.69	11.70
Log hours worked	5.04	4.95

Notes: Figures are means (continuous variables) or percentages (categorical variables), pooled across waves 1–3 using available non-missing observations. The restricted sample includes only individuals with complete earnings and sectoral-choice information in all three waves (4,534 individuals; 13,602 person-waves); the unrestricted sample pools all available person-wave observations (up to 43,715).

Table 6: Descriptive Statistics for Individuals – Part A

Variable	Number of Survey Round 1	2	3	Total
N	4,534	4,534	4,534	13,602
	(33.3%)	(33.3%)	(33.3%)	(100.0%)
hh_count	3,894	3,922	3,943	11,759
	(98.8%)	(99.5%)	(100.0%)	(100.0%)
gender	0.724	0.724	0.724	0.724
	(0.447)	(0.447)	(0.447)	(0.447)
age	40.298	44.229	47.197	43.908
	(11.198)	(11.121)	(11.176)	(11.516)
non-agriculture	0.661	0.661	0.637	0.653
	(0.473)	(0.473)	(0.481)	(0.476)
wagedwork	0.456	0.449	0.414	0.440
	(0.498)	(0.497)	(0.493)	(0.496)
income	125,697.765	206,341.100	449,344.429	260,461.100
	(161,273.500)	(284,544.700)	(1,133,006.392)	(694,561.000)
ln_income	11.135	11.632	12.312	11.693
	(1.203)	(1.205)	(1.230)	(1.305)
hours worked	178.170	171.850	173.064	174.375
	(74.288)	(75.045)	(81.763)	(77.157)
ln_hours worked	5.078	5.028	5.024	5.043
	(0.503)	(0.545)	(0.569)	(0.540)
cpi	145.207	194.363	209.455	183.008
	(4.605)	(7.083)	(9.442)	(28.390)
ln_cpi	4.978	5.269	5.343	5.197
	(0.031)	(0.036)	(0.045)	(0.162)
ruralborn	0.744	0.747	0.790	0.761
	(0.436)	(0.435)	(0.407)	(0.427)
moved	0.533	0.467	0.464	0.487
	(0.499)	(0.499)	(0.499)	(0.500)
urban	0.445	0.441	0.439	0.442
	(0.497)	(0.497)	(0.496)	(0.497)

Notes: Top line reports means (or counts for N, *hh_count*); second line reports standard deviations for continuous/binary variables and percentages for N and *hh_count*. Monetary values in Indonesian Rupiah. Statistics use the balanced panel (waves 1–3).

Table 7: Descriptive Statistics for Individuals – Part B (Categorical Variables)

	Number of Survey Round			
Category	**1**	**2**	**3**	**Total**
N	4,534 (33.3%)	4,534 (33.3%)	4,534 (33.3%)	13,602 (100.0%)
MARITAL STATUS				
Not yet married	97 (2.1%)	53 (1.2%)	37 (0.8%)	187 (1.4%)
Married	4,159 (91.7%)	4,120 (90.9%)	4,072 (89.8%)	12,351 (90.8%)
Separated	17 (0.4%)	22 (0.5%)	20 (0.4%)	59 (0.4%)
Divorced	56 (1.2%)	54 (1.2%)	69 (1.5%)	179 (1.3%)
Widowed	205 (4.5%)	285 (6.3%)	336 (7.4%)	826 (6.1%)
EDUCATION				
Unschooled	679 (15.0%)	636 (14.0%)	603 (13.3%)	1,918 (14.1%)
Primary	2,327 (51.4%)	2,385 (52.6%)	2,316 (51.1%)	7,028 (51.7%)
Junior high	541 (11.9%)	501 (11.1%)	477 (10.5%)	1,519 (11.2%)
Senior high	753 (16.6%)	738 (16.3%)	660 (14.6%)	2,151 (15.8%)
College/University	231 (5.1%)	271 (6.0%)	339 (7.5%)	841 (6.2%)
Others	0 (0.0%)	2 (0.0%)	134 (3.0%)	136 (1.0%)
RELIGION				
Islam	3,932 (86.7%)	3,954 (87.2%)	3,944 (87.0%)	11,830 (87.0%)
Protestant	186 (4.1%)	184 (4.1%)	191 (4.2%)	561 (4.1%)
Catholic	90 (2.0%)	90 (2.0%)	96 (2.1%)	276 (2.0%)
Hindu	283 (6.2%)	278 (6.1%)	281 (6.2%)	842 (6.2%)
Buddhist	22 (0.5%)	21 (0.5%)	18 (0.4%)	61 (0.5%)
Others	21 (0.5%)	7 (0.2%)	4 (0.1%)	32 (0.2%)

Notes: Each cell shows count with share in parentheses. Shares sum to 100% within each block.

Table 8: Descriptive Statistics for Households

	Number of Survey Round			
Variable	**1**	**2**	**3**	**Total**
N	3,894	3,922	3,943	11,759
	(33.1%)	(33.4%)	(33.5%)	(100.0%)
panel	.	0.979	0.978	0.979
	(.)	(0.142)	(0.147)	(0.145)
household size	4.777	5.357	5.794	5.311
	(1.975)	(2.172)	(2.401)	(2.229)
urban	0.432	0.428	0.430	0.430
	(0.495)	(0.495)	(0.495)	(0.495)
own farm business	0.430	0.394	0.470	0.432
	(0.495)	(0.489)	(0.499)	(0.495)
farm business assets	7,010,372.467	10,295,959.316	26,326,949.171	15,570,080.897
	(34,058,554.565)	(21,197,639.074)	(83,593,608.857)	(57,621,680.231)
ln_farm business assets	14.054	14.785	15.546	14.855
	(2.215)	(2.010)	(2.124)	(2.210)
owns non-farm business	0.376	0.400	0.525	0.434
	(0.484)	(0.490)	(0.499)	(0.496)
non-farm business assets	6,355,246.787	5,254,494.918	9,212,882.056	7,165,417.153
	(60,065,365.723)	(24,286,231.723)	(43,900,334.760)	(44,321,580.103)
ln_non-farm business assets	12.249	12.955	13.345	12.920
	(2.349)	(2.335)	(2.429)	(2.418)
own family business	0.673	0.671	0.800	0.715
	(0.469)	(0.470)	(0.400)	(0.451)
own assets	0.976	0.999	0.999	0.991
	(0.154)	(0.036)	(0.036)	(0.094)
total assets not for business	13,823,547.269	20,289,255.751	37,518,475.950	24,002,955.091
	(77,331,870.416)	(52,625,303.361)	(86,228,415.877)	(74,118,228.152)
ln_total assets not for business	14.711	15.654	16.392	15.596
	(1.779)	(1.622)	(1.587)	(1.799)
real estate not for business	11,080,223.087	18,572,157.272	32,119,842.889	20,931,444.714
	(54,346,295.877)	(49,455,894.231)	(67,942,887.649)	(58,598,548.269)
ln_real estate not for business	14.825	15.624	16.311	15.612
	(1.571)	(1.519)	(1.522)	(1.651)
shock	0.313	0.401	0.341	0.352
	(0.464)	(0.490)	(0.474)	(0.478)
numbers of shock	0.394	0.564	0.435	0.464
	(0.653)	(0.821)	(0.692)	(0.729)

Notes: Top line reports means (or counts for N); second line reports standard deviations (or column shares for N). Monetary values are in Indonesian Rupiah.

Table 9: SUR: Outcome Variable, log Earnings in 1993

ln_earnings_1	**(1)**	**(2)**	**(3)**	**(4)**
nonag_1	0.654 **	0.555 **	0.509 **	0.516 **
	(0.092)	(0.091)	(0.081)	(0.081)
nonag_2	-0.036	-0.058	-0.105	-0.098
	(0.101)	(0.099)	(0.090)	(0.089)
nonag_3	-0.045	-0.065	-0.015	0.009
	(0.096)	(0.094)	(0.084)	(0.083)
nonag_1 * nonag_2	0.058	0.076	0.090	0.116
	(0.151)	(0.148)	(0.133)	(0.132)
nonag_1 * nonag_3	0.331 *	0.303 *	0.139	0.116
	(0.170)	(0.167)	(0.150)	(0.149)
nonag_2 * nonag_3	0.264	0.250	0.183	0.162
	(0.166)	(0.163)	(0.146)	(0.145)
nonag_1 * nonag_2 * nonag_3	-0.087	-0.192	-0.118	-0.118
	(0.231)	(0.227)	(0.203)	(0.202)
urban_1		0.453 **	0.189 **	0.188 **
		(0.098)	(0.087)	(0.087)
urban_2		-0.030	0.129	0.091
		(0.115)	(0.103)	(0.102)
urban_3		0.071	-0.055	-0.039
		(0.084)	(0.076)	(0.075)
ln_hrsworked_1			0.364 **	0.356 **
			(0.031)	(0.031)
ln_hrsworked_2			0.057 **	0.054 *
			(0.028)	(0.028)
ln_hrsworked_3			0.031	0.034
			(0.027)	(0.027)
wagedwork_1			0.110 **	0.111 **
			(0.042)	(0.042)
wagedwork_2			0.099 **	0.099 **
			(0.045)	(0.045)
wagedwork_3			-0.077 *	-0.070 *
			(0.042)	(0.042)
ln_cpi_1				-3.172 **
				(0.946)
ln_cpi_2				3.678 **
				(0.730)
ln_cpi_3				1.060 **
				(0.411)
shock_1				-0.025
				(0.031)
shock_2				-0.009
				(0.030)
shock_3				0.043
				(0.031)
province	N	Y	Y	N
age	N	N	Y **	Y **
gender	N	N	Y **	Y **
marital_status	N	N	Y	Y
religion	N	N	Y **	Y **
education	N	N	Y **	Y **
constant	10.423 **	10.408 **	6.922 **	-2.342
	(0.032)	(0.050)	(0.230)	(3.768)
N	4,534	4,533	4,428	4,425
R^2	0.188	0.220	0.392	0.402

Notes: Numbers in parentheses are **standard errors**. Asterisks indicate significance: ***** $p < 0.10$, ****** $p < 0.05$.

Table 10: SUR: Outcome Variable, log Earnings in 1997

ln_earnings_2	**(1)**	**(2)**	**(3)**	**(4)**
nonag_1	0.273 **	0.185 **	0.182 **	0.191 **
	(0.092)	(0.091)	(0.080)	(0.079)
nonag_2	0.453 **	0.430 **	0.331 **	0.346 **
	(0.101)	(0.100)	(0.089)	(0.088)
nonag_3	-0.078	-0.097	-0.099	-0.082
	(0.096)	(0.094)	(0.083)	(0.082)
nonag_1 * nonag_2	-0.076	-0.058	-0.055	-0.038
	(0.152)	(0.149)	(0.132)	(0.131)
nonag_1 * nonag_3	0.332 *	0.311 *	0.157	0.131
	(0.171)	(0.168)	(0.148)	(0.146)
nonag_2 * nonag_3	0.151	0.145	0.080	0.055
	(0.166)	(0.164)	(0.145)	(0.143)
nonag_1 * nonag_2 * nonag_3	0.111	0.006	0.079	0.096
	(0.231)	(0.228)	(0.201)	(0.199)
urban_1		0.438 **	0.181 **	0.197 **
		(0.098)	(0.086)	(0.085)
urban_2		-0.131	0.037	0.027
		(0.115)	(0.102)	(0.100)
urban_3		0.131	0.007	-0.032
		(0.084)	(0.075)	(0.074)
ln_hrsworked_1			0.095 **	0.084 **
			(0.031)	(0.030)
ln_hrsworked_2			0.272 **	0.272 **
			(0.028)	(0.028)
ln_hrsworked_3			0.067 **	0.070 **
			(0.027)	(0.027)
wagedwork_1			-0.019	-0.007
			(0.042)	(0.041)
wagedwork_2			0.192 **	0.188 **
			(0.044)	(0.044)
wagedwork_3			-0.022	-0.015
			(0.042)	(0.042)
ln_cpi_1				-5.389 **
				(0.933)
ln_cpi_2				4.841 **
				(0.720)
ln_cpi_3				0.479
				(0.405)
shock_1				0.068 **
				(0.031)
shock_2				-0.045
				(0.030)
shock_3				0.005
				(0.030)
province	N	Y **	Y **	N
age	N	N	Y **	Y **
gender	N	N	Y **	Y **
marital_status	N	N	Y	Y
religion	N	N	Y **	Y **
education	N	N	Y **	Y **
constant	10.907 **	10.990 **	7.851 **	6.500 *
	(0.032)	(0.051)	(0.227)	(3.715)
N	4,534	4,533	4,428	4,425
R^2	0.187	0.214	0.411	0.422

Notes: Numbers in parentheses are **standard errors**. Asterisks indicate significance: ***** $p < 0.10$, ****** $p < 0.05$.

Table 11: SUR: Outcome Variable, log Earnings in 2000

ln_earnings_3	**(1)**	**(2)**	**(3)**	**(4)**
nonag_1	0.237 **	0.165 *	0.146 *	0.158 *
	(0.099)	(0.098)	(0.088)	(0.088)
nonag_2	0.005	-0.019	-0.159	-0.143
	(0.108)	(0.107)	(0.098)	(0.098)
nonag_3	0.191 *	0.174 *	0.156 *	0.165 *
	(0.103)	(0.102)	(0.091)	(0.091)
nonag_1 * nonag_2	-0.282 *	-0.260	-0.132	-0.125
	(0.162)	(0.161)	(0.145)	(0.144)
nonag_1 * nonag_3	0.204	0.189	0.054	0.038
	(0.183)	(0.181)	(0.163)	(0.162)
nonag_2 * nonag_3	0.171	0.172	0.169	0.151
	(0.178)	(0.176)	(0.159)	(0.159)
nonag_1 * nonag_2 * nonag_3	0.332	0.240	0.189	0.201
	(0.248)	(0.246)	(0.221)	(0.220)
urban_1		0.411 **	0.141	0.160 *
		(0.106)	(0.095)	(0.095)
urban_2		-0.329 **	-0.127	-0.148
		(0.124)	(0.112)	(0.111)
urban_3		0.263 **	0.139 *	0.121
		(0.091)	(0.082)	(0.081)
ln_hrsworked_1			0.006	-0.003
			(0.034)	(0.034)
ln_hrsworked_2			0.104 **	0.107 **
			(0.031)	(0.031)
ln_hrsworked_3			0.257 **	0.259 **
			(0.030)	(0.029)
wagedwork_1			0.013	0.026
			(0.046)	(0.046)
wagedwork_2			0.064	0.063
			(0.049)	(0.049)
wagedwork_3			-0.007	-0.005
			(0.046)	(0.046)
ln_cpi_1				-3.409 **
				(1.032)
ln_cpi_2				2.717 **
				(0.796)
ln_cpi_3				0.634
				(0.448)
shock_1				0.062 *
				(0.034)
shock_2				0.020
				(0.033)
shock_3				-0.043
				(0.034)
province	N	Y **	Y	N
age	N	N	Y **	Y **
gender	N	N	Y **	Y **
marital_status	N	N	Y	Y
religion	N	N	Y	Y *
education	N	N	Y **	Y **
constant	11.800 **	11.877 **	9.248 **	8.428 **
	(0.035)	(0.055)	(0.250)	(4.110)
N	4,534	4,533	4,428	4,425
R^2	0.106	0.124	0.314	0.319

Notes: Numbers in parentheses are **standard errors**. Asterisks indicate significance: ***** $p < 0.10$, ****** $p < 0.05$.

Table 12: Stayers and Switchers

	Ag stayer	Ag switcher	Nonag switcher	Nonag stayer
N (% of sample)	1,137 (25.1)	401 (8.8)	474 (10.5)	2,522 (55.6)
Panel A. Continuous variables (mean, SD)				
Age (years)	44.7 (12.2)	40.0 (11.4)	41.6 (11.2)	38.1 (10.1)
Log earnings	10.42 (1.17)	10.45 (1.28)	11.14 (1.12)	11.56 (1.00)
Log hours worked	5.02 (0.44)	4.96 (0.53)	5.08 (0.55)	5.12 (0.51)
Panel B. Education (% of group)				
Unschooled	25.1	21.2	18.1	8.8
Primary school	63.1	59.9	59.1	43.2
Junior high	6.6	9.2	8.4	15.4
Senior high	4.6	8.7	11.0	24.4
College / university	0.6	1.0	3.4	8.1
Panel C. Other characteristics (% of group)				
Female	12.2	17.7	19.6	37.6
Waged work	21.3	34.2	52.5	57.1
Urban	11.6	18.0	33.3	65.6

Notes: A *stayer* remains in the same sector in all three waves; a *switcher* changes sector at least once; groups are labelled by their 1993 sector. All characteristics are measured in 1993. Panel A reports means with standard deviations in parentheses; Panels B and C report shares within each group. Balanced panel, $N = 4{,}534$.

Table 13: Robustness SUR: Outcome Variable, Log Earnings in 1993

ln_earnings_1	**(i)**	**(ii)**	**(iii)**	**(iv)**
nonag_1	0.516 **	0.508 **	0.538 **	0.480 **
	(0.081)	(0.078)	(0.079)	(0.078)
nonag_2	-0.098	-0.089	-0.093	-0.138
	(0.089)	(0.087)	(0.088)	(0.088)
nonag_3	0.009	0.034	0.059	0.027
	(0.083)	(0.081)	(0.082)	(0.082)
nonag_1 * nonag_2	0.116	0.113	0.102	0.124
	(0.132)	(0.129)	(0.130)	(0.129)
nonag_1 * nonag_3	0.116	0.081	0.052	0.005
	(0.149)	(0.145)	(0.148)	(0.145)
nonag_2 * nonag_3	0.162	0.132	0.115	0.108
	(0.145)	(0.142)	(0.143)	(0.143)
nonag_1 * nonag_2 * nonag_3	-0.118	-0.083	-0.060	0.001
	(0.202)	(0.196)	(0.200)	(0.197)
urban_1	0.188 **	0.191 **	0.185 **	0.149 *
	(0.087)	(0.084)	(0.087)	(0.085)
urban_2	0.091	0.083	0.109	0.162
	(0.102)	(0.099)	(0.102)	(0.100)
urban_3	-0.039	-0.035	-0.047	-0.130 *
	(0.075)	(0.073)	(0.075)	(0.073)
ln_hrsworked_1	0.356 **	0.340 **	0.357 **	0.356 **
	(0.031)	(0.030)	(0.031)	(0.030)
ln_hrsworked_2	0.054 *	0.057 **	0.048 *	0.033
	(0.028)	(0.027)	(0.028)	(0.027)
ln_hrsworked_3	0.034	0.035	0.029	0.038
	(0.027)	(0.026)	(0.027)	(0.027)
wagedwork_1	0.111 **	0.111 **	0.104 **	0.112 **
	(0.042)	(0.041)	(0.042)	(0.044)
wagedwork_2	0.099 **	0.108 **	0.087 **	0.120 **
	(0.045)	(0.044)	(0.044)	(0.046)
wagedwork_3	-0.070 *	-0.068 *	-0.059	-0.034
	(0.042)	(0.041)	(0.042)	(0.044)
familybiz_1				-0.051
				(0.040)
familybiz_2				-0.005
				(0.040)
familybiz_3				-0.015
				(0.044)
ln_assets_1				0.077 **
				(0.010)
ln_assets_2				0.067 **
				(0.012)
ln_assets_3				0.068 **
				(0.012)
CPI	Y	Y	Y	Y
shock	Y	Y	Y	Y
age	Y	Y	Y	Y
gender	Y	Y	Y	Y
marital_status	Y	Y	Y	Y
religion	Y	Y	Y	Y
education	Y	Y	Y	Y
constant	-2.342	-2.249	-2.542	-3.808
	(3.768)	(3.669)	(3.758)	(3.679)
N	4,425	4,425	4,505	4,302
R^2	0.402	0.407	0.401	0.445

Notes: Numbers in parentheses are **standard errors**. Asterisks indicate significance: ***** $p < 0.10$, ****** $p < 0.05$.

Table 14: Robustness SUR: Outcome Variable, Log Earnings in 1997

ln_earnings_2	**(i)**	**(ii)**	**(iii)**	**(iv)**
nonag_1	0.191 **	0.233 **	0.180 **	0.138 *
	(0.079)	(0.076)	(0.078)	(0.077)
nonag_2	0.346 **	0.334 **	0.353 **	0.310 **
	(0.088)	(0.085)	(0.086)	(0.086)
nonag_3	-0.082	-0.079	-0.054	-0.048
	(0.082)	(0.079)	(0.081)	(0.080)
nonag_1 * nonag_2	-0.038	-0.083	-0.021	-0.018
	(0.131)	(0.125)	(0.128)	(0.127)
nonag_1 * nonag_3	0.131	0.078	0.110	0.010
	(0.146)	(0.141)	(0.146)	(0.142)
nonag_2 * nonag_3	0.055	0.059	0.027	-0.019
	(0.143)	(0.138)	(0.142)	(0.140)
nonag_1 * nonag_2 * nonag_3	0.096	0.142	0.099	0.235
	(0.199)	(0.191)	(0.197)	(0.193)
urban_1	0.197 **	0.194 **	0.198 **	0.138 *
	(0.085)	(0.082)	(0.086)	(0.083)
urban_2	0.027	0.024	0.040	0.126
	(0.100)	(0.096)	(0.100)	(0.097)
urban_3	-0.032	-0.025	-0.043	-0.129 *
	(0.074)	(0.071)	(0.074)	(0.071)
ln_hrsworked_1	0.084 **	0.079 **	0.084 **	0.088 **
	(0.030)	(0.029)	(0.030)	(0.029)
ln_hrsworked_2	0.272 **	0.269 **	0.265 **	0.246 **
	(0.028)	(0.027)	(0.028)	(0.027)
ln_hrsworked_3	0.070 **	0.068 **	0.075 **	0.065 **
	(0.027)	(0.026)	(0.026)	(0.026)
wagedwork_1	-0.007	0.013	-0.010	0.015
	(0.041)	(0.040)	(0.041)	(0.043)
wagedwork_2	0.188 **	0.173 **	0.181 **	0.204 **
	(0.044)	(0.042)	(0.044)	(0.045)
wagedwork_3	-0.015	-0.011	-0.015	0.016
	(0.042)	(0.040)	(0.041)	(0.043)
familybiz_1				0.002
				(0.039)
familybiz_2				-0.034
				(0.039)
familybiz_3				-0.008
				(0.043)
ln_assets_1				0.010
				(0.010)
ln_assets_2				0.139 **
				(0.012)
ln_assets_3				0.076 **
				(0.012)
CPI	Y	Y	Y	Y
shock	Y	Y	Y	Y
age	Y	Y	Y	Y
gender	Y	Y	Y	Y
marital_status	Y	Y	Y	Y
religion	Y	Y	Y	Y
education	Y	Y	Y	Y
constant	6.500 *	6.574 *	6.938 *	5.508
	(3.715)	(3.564)	(3.708)	(3.597)
N	4,425	4,425	4,505	4,302
R^2	0.422	0.434	0.419	0.472

Notes: Numbers in parentheses are **standard errors**. Asterisks indicate significance: ***** $p < 0.10$, ****** $p < 0.05$.

Table 15: Robustness SUR: Outcome Variable, Log Earnings in 2000

ln_earnings_3	**(i)**	**(ii)**	**(iii)**	**(iv)**
nonag_1	0.158 *	0.159 *	0.154 *	0.112
	(0.088)	(0.084)	(0.086)	(0.085)
nonag_2	-0.143	-0.146	-0.155	-0.163 *
	(0.098)	(0.093)	(0.095)	(0.095)
nonag_3	0.165 *	0.146 *	0.212 **	0.162 *
	(0.091)	(0.086)	(0.089)	(0.089)
nonag_1 * nonag_2	-0.125	-0.133	-0.105	-0.154
	(0.144)	(0.137)	(0.141)	(0.140)
nonag_1 * nonag_3	0.038	0.011	-0.003	-0.053
	(0.162)	(0.154)	(0.160)	(0.157)
nonag_2 * nonag_3	0.151	0.156	0.122	0.094
	(0.159)	(0.151)	(0.156)	(0.155)
nonag_1 * nonag_2 * nonag_3	0.201	0.238	0.223	0.343
	(0.220)	(0.209)	(0.217)	(0.213)
urban_1	0.160 *	0.164 *	0.154	0.110
	(0.095)	(0.090)	(0.094)	(0.092)
urban_2	-0.148	-0.145	-0.133	-0.058
	(0.111)	(0.106)	(0.111)	(0.108)
urban_3	0.121	0.109	0.117	0.009
	(0.081)	(0.077)	(0.081)	(0.079)
ln_hrsworked_1	-0.003	-0.013	-0.002	-0.001
	(0.034)	(0.032)	(0.033)	(0.033)
ln_hrsworked_2	0.107 **	0.115 **	0.097 **	0.088 **
	(0.031)	(0.029)	(0.030)	(0.030)
ln_hrsworked_3	0.259 **	0.249 **	0.262 **	0.264 **
	(0.029)	(0.028)	(0.029)	(0.029)
wagedwork_1	0.026	0.023	0.017	0.078 *
	(0.046)	(0.044)	(0.045)	(0.048)
wagedwork_2	0.063	0.061	0.059	0.078
	(0.049)	(0.046)	(0.048)	(0.050)
wagedwork_3	-0.005	0.010	0.003	0.004
	(0.046)	(0.044)	(0.045)	(0.048)
familybiz_1				0.026
				(0.043)
familybiz_2				0.021
				(0.043)
familybiz_3				-0.105 **
				(0.047)
ln_assets_1				0.024 **
				(0.011)
ln_assets_2				0.084 **
				(0.013)
ln_assets_3				0.147 **
				(0.013)
CPI	Y	Y	Y	Y
shock	Y	Y	Y	Y
age	Y	Y	Y	Y
gender	Y	Y	Y	Y
marital_status	Y	Y	Y	Y
religion	Y	Y	Y	Y
education	Y	Y	Y	Y
constant	8.428 **	7.978 **	8.774 **	7.602 *
	(4.110)	(3.909)	(4.083)	(3.985)
N	4,425	4,425	4,505	4,302
R^2	0.319	0.335	0.320	0.377

Notes: Numbers in parentheses are **standard errors**. Asterisks indicate significance: ***** $p < 0.10$, ****** $p < 0.05$.

Table 16: Robustness Structural Parameters

Structural Parameters	**(i)**		**(ii)**		**(iii)**		**(iv)**	
λ_1	0.174	**	0.197	**	0.172	**	0.125	**
	(0.053)		(0.051)		(0.052)		(0.053)	
λ_2	-0.083		-0.083		-0.087		-0.114	**
	(0.058)		(0.056)		(0.057)		(0.056)	
λ_3	-0.084		-0.078		-0.046		-0.066	
	(0.053)		(0.051)		(0.051)		(0.050)	
λ_4	-0.010		-0.034		-0.003		-0.010	
	(0.080)		(0.079)		(0.079)		(0.076)	
λ_5	0.102		0.064		0.059		-0.010	
	(0.095)		(0.092)		(0.094)		(0.085)	
λ_6	0.124		0.119		0.088		0.058	
	(0.088)		(0.087)		(0.086)		(0.084)	
λ_7	0.060		0.117		0.087		0.189	
	(0.134)		(0.140)		(0.134)		(0.150)	
α	0.350	**	0.324	**	0.363	**	0.359	**
	(0.064)		(0.065)		(0.062)		(0.065)	
β	-0.053		-0.117		-0.050		-0.012	
	(0.376)		(0.343)		(0.397)		(0.466)	

Notes: Columns (i)–(iv) report the structural parameters corresponding to first-stage specifications (i)–(iv) in the Robustness SUR tables, respectively. Numbers in parentheses are standard errors; ****** $p < 0.05$, ***** $p < 0.10$.

9 Appendix

9.1 Background on Suri's Empirical Approach

This appendix provides background on the empirical approach of Suri (2011), which this paper adapts to study agricultural productivity gaps. While the main text explains how the framework is modified for the APG context, this appendix reviews Suri's original application to technology adoption in Kenya and outlines the underlying logic of the correlated random coefficient (CRC) model. The purpose is to provide readers less familiar with this method with a clear understanding of its origins, intuition, and technical foundations. Readers already acquainted with Suri's work may skip directly to the empirical-model section, where the adapted framework is presented.

The CRC model is well suited to the two challenges this study must overcome—measuring sector-specific unobserved ability and disentangling the endogeneity of sector choice. It recovers individual sorting on unobserved ability by exploiting choice histories, without imposing parametric distributional assumptions on latent heterogeneity.

In Suri's empirical strategy, expected potential returns determine each farmer's adoption decision on hybrid seeds, which follows Heckman and Vytlacil (1998) under the generalized Roy's model framework (Roy, 1951). Moreover, the method imposes no functional form on unobserved heterogeneity: it exploits the fact that a farmer's adoption choice history reveals her net benefits from the technology, in the spirit of Chamberlain's panel fixed-effect estimation (Chamberlain, 1982, 1984). This framework consists of two key appealing features: First, it considers each farmer's net benefit as a deviation from the average net benefit of hybrid seed adoption, which explicitly models heterogeneity. Second, it exploits the revealed comparative advantages that can be projected by hybrid seeds adoption trajectories, which allows for estimating the selection effect without distributional assumptions. As a result, Suri's (2011) empirical approach is preferable for tackling the two challenges that the research question of this paper must overcome.

9.2 Decomposing APG

This appendix presents the step-by-step derivation of the APG components described in the main text, as given by equations (31) and (32). It starts by defining APG as the average difference in earnings between two sectors, illustrated in Section 3.4. Then, it uses the specified empirical model for the observed earnings and aggregates it to obtain an algebraic expression for the APG. Finally, it breaks down the key components of the APG to provide additional insights into unobserved information in the data. The object of interest is S_θ, that is, the component of the APG arising from the sorting into the sector based on unobserved comparative advantage. Also, d_θ is meaningful as it captures the sector-composition difference in unobserved comparative advantage that explains the APG.

The APG defined in equation (30) can be expressed as

$$APG = \underbrace{E[w_i \mid D_i = 1]}_{\text{avg. log earnings in non-ag}} - \underbrace{E[w_i \mid D_i = 0]}_{\text{avg. log earnings in ag}} \tag{49}$$

Recall the observed log earnings (suppressing t for ease of exposition):

$$w_i = \delta^a + (\delta^n - \delta^a)D_i \; + \; \tau_i + \theta_i + \beta\,\theta_i D_i \; + \; X_i\gamma^a + X_i(\gamma^n - \gamma^a)D_i \; + \; \epsilon_i, \tag{50}$$

where $D_i \in \{0, 1\}$ indicates non-agriculture, τ_i is individual latent ability that is irrelevant to the sector choice, θ_i is unobserved comparative advantage that individuals take into account when choosing sectors, and β is the *relative* loading of non-agriculture on θ_i, which is the selection effect defined by the main model in this chapter.

Then, the average log earnings in non-agriculture and agriculture are given by (51) and (52).

$$\begin{aligned} E[w_i \mid D_i = 1] = \delta^n \; + \; &\underbrace{E[\tau_i \mid D_i = 1]}_{\equiv \mu_\tau^n} \\ &+ \; (1+\beta)\, E[\theta_i \mid D_i = 1] \; + \; \gamma^n E[X_i \mid D_i = 1] + \underbrace{E[\epsilon_i^n \mid D_i = 1]}_{=0}, \end{aligned} \tag{51}$$

$$\begin{aligned} E[w_i \mid D_i = 0] = \delta^a \; + \; &\underbrace{E[\tau_i \mid D_i = 0]}_{\equiv \mu_\tau^a} \\ &+ \; E[\theta_i \mid D_i = 0] \; + \; \gamma^a E[X_i \mid D_i = 0] + \underbrace{E[\epsilon_i^a \mid D_i = 0]}_{=0}. \end{aligned} \tag{52}$$

As ϵ_i^n and ϵ_i^a are transitory errors, their conditional means are zero and drop out. Define $\mu_\tau^n \equiv E[\tau_i|D_i = 1]$ and $\mu_\tau^a \equiv E[\tau_i|D_i = 0]$ as the average sector-neutral latent ability among non-agricultural and agricultural workers, respectively. For a given individual, τ_i enters both her potential earnings identically (equations (18)–(19)), so it cancels in the potential-earnings comparison that governs her sector choice. For instance, if work ethic rewards an individual the same in potential earnings across sectors, then it is not something she chooses a sector on. The two group averages in the APG, however, are taken over different individuals—those who actually work in non-agriculture versus those who actually farm. So μ_τ^n and μ_τ^a need not be equal. Their difference reflects workforce composition, not how τ_i is rewarded across sectors. Subtracting (52) from (51) gives equation (53).

$$\begin{aligned} APG = & \underbrace{(\delta^n - \delta^a)}_{\equiv \Delta_\delta} + \underbrace{\gamma^n E[X_i \mid D = 1] - \gamma^a E[X_i \mid D = 0]}_{\equiv \Delta_X} \\ & + \underbrace{\beta\, E[\theta_i \mid D = 1] \;+\; \big(E[\theta_i \mid D = 1] - E[\theta_i \mid D = 0]\big) \;+\; \big(E[\tau_i \mid D = 1] - E[\tau_i \mid D = 0]\big)}_{\equiv \Delta_\theta} \end{aligned} \tag{53}$$

Define the average sector-wide efficiency gap as Δ_δ:

$$\Delta_\delta \equiv \delta^n - \delta^a \tag{54}$$

Let Δ_X be the difference in the average returns from observed characteristics between two groups:

$$\Delta_X \equiv \gamma^n E[X_i \mid D = 1] - \gamma^a E[X_i \mid D = 0] \tag{55}$$

Where Δ_θ denotes the average difference in rewards between workers and farmers due to their latent abilities across sectors:

$$\begin{aligned} \Delta_\theta \equiv & \underbrace{\beta\, E[\theta_i \mid D = 1]}_{\equiv S_\theta} \;+\; \underbrace{\big(E[\theta_i \mid D = 1] - E[\theta_i \mid D = 0]\big)}_{\equiv d_\theta} \\ & + \; \underbrace{\big(E[\tau_i \mid D = 1] - E[\tau_i \mid D = 0]\big)}_{\equiv d_\tau} \end{aligned} \tag{56}$$

Therefore, the APG consists of three components: the sector-wide efficiency gap (Δ_δ), the average earnings disparity due to observed characteristics (Δ_X), and the differential rewards arising from unobserved absolute advantages between sectors (Δ_θ).

$$APG = \Delta_\delta + \Delta_X + \Delta_\theta \tag{31}$$

Furthermore, Δ_θ can be divided into three parts, as presented in equation (32): (i) the selection term, that is, the mean differential rewards for the non-agricultural workers when comparing what rewards they would have had if they had worked in the agricultural sector, given by S_θ; (ii) the group composition, which is, the gap in unobserved comparative advantage between the non-agricultural and agricultural sectors, denoted by d_θ; and (iii) the group difference in the latent abilities that are irrelevant to sectoral choices, d_τ.

$$\Delta_\theta = \underbrace{\beta E[\theta_i \mid D_i = 1]}_{\equiv S_\theta} + \underbrace{E[\theta_i \mid D_i = 1] - E[\theta_i \mid D_i = 0]}_{\equiv d_\theta} + d_\tau \tag{32}$$

As shown in Section 3.2, θ_i^n and θ_i^a are unobserved absolute advantages in non-agriculture and agriculture, respectively. They can be rewritten as a function of unobserved comparative advantage θ_i and latent time-invariant skills τ_i. Hence, S_θ and d_θ are the objects of interest; the former is the selection term, and the latter is the composition effect arising from unobserved comparative advantages. Hence, I will present how those two explain the APG, $\frac{S_\theta}{APG}$ and $\frac{d_\theta}{APG}$, respectively.

In sum, this appendix shows that APG can be divided into three components: the sector-wide efficiency gap (Δ_δ), the observed earnings disparity (Δ_X), and the differential returns to latent abilities (Δ_θ). Moreover, within Δ_θ, the objects of interest are the selection term S_θ and the composition term d_θ. Specifically, S_θ/APG answers the extent to which sorting contributes to the APG.

9.3 Recovering Structural Parameters

This appendix is to demonstrate how the structure parameter, unobserved comparative advantage, θ_i, is recovered in a simplified two-period no-covariant model. θ_i is unobserved in the data. I will now demonstrate how the structural parameters of interest can be recovered without imposing distributional assumptions. Following the procedure proposed by Suri (2011), I present the recovery of structural parameters in the simplest setting, without covariates, over two periods, as expressed in equation (57).

$$w_{it} = \eta + \alpha D_{it} + \theta_i + \beta\theta_i D_{it} + u_{it} \tag{57}$$

where $\delta_t^a = \eta \quad \forall t$, $\alpha \equiv \delta_t^n - \delta_t^a \quad \forall t$, and $u_{it} \equiv \tau_i + \epsilon_{it}$.

I can do this because structural parameters β and θ_i do not enter covariates in the main estimation equation (23). First, I disentangle the dependency between θ_i and D_{it} by linearly projecting θ_i onto the entire history of sector choices and their interaction terms. This method was first developed by Chamberlain (Chamberlain, 1982, 1984) to estimate individual unobserved fixed effects in panel data. Later, Suri (2011) generalized Chamberlain's fixed-effect estimation by including interactions of choice histories to purge the dependency between unobserved abilities and individual choices fully. Chamberlain and Suri treat this as a purely technical step to separate θ_i into two parts: one is related to sectoral choice (D_{it}), and the remainder is orthogonal to the sectoral choices. Since the choice variable D_{it} is a dummy variable, the projection θ_i onto a complete history and interaction terms will be a saturated model to purge the correlation between θ_i and D_{it}, which is formally expressed in the equation (58). However, I interpret the equation (58) as individual choice trajectories that reveal the unobserved comparative advantages, θ_i.

$$\theta_i = \lambda_0 + \lambda_1 D_{i1} + \lambda_2 D_{i2} + \lambda_3 D_{i1} D_{i2} + \nu_i \tag{58}$$

Next, I substitute equation (58) into the wage equation (57) to obtain the log earnings for each period as a function of choice histories and their interactions, see equations (59) and (60).

$$\begin{aligned} w_{i1} = {} & (\eta + \lambda_0) + (\lambda_1(1+\beta) + \alpha + \beta\lambda_0) D_{i1} \\ & + \lambda_2 D_{i2} + (\lambda_3(1+\beta) + \beta\lambda_2) D_{i1} D_{i2} + (\nu_i + \beta\nu_i D_{i1} + u_{i1}) \end{aligned} \tag{59}$$

$$\begin{aligned} w_{i2} = {} & (\eta + \lambda_0) + \lambda_1 D_{i1} \\ & + (\lambda_2(1+\beta) + \alpha + \beta\lambda_0) D_{i2} + (\lambda_3(1+\beta) + \beta\lambda_1) D_{i1} D_{i2} \\ & + (\nu_i + \beta\nu_i D_{i2} + u_{i2}) \end{aligned} \tag{60}$$

Since sector choices are observed in each period, I can run a reduced-form regression of earnings on the choice history and their interactions in this stacked system of equations (59) and (60). To simplify the coefficients in the reduced form regression in equations (59) and (60), I can rewrite them as equations (61) and (62).

$$w_{i1} = \eta_1 + \phi_1 D_{i1} + \phi_2 D_{i2} + \phi_3 D_{i1} D_{i2} + e_{i1} \tag{61}$$

$$w_{i2} = \eta_2 + \phi_4 D_{i1} + \phi_5 D_{i2} + \phi_6 D_{i1} D_{i2} + e_{i2} \tag{62}$$

The reduced form regression of each period earnings on the entire history of the sector choice, including interaction terms across periods, will obtain reduced form coefficients ϕ_1, ϕ_2, ϕ_3, ϕ_4, ϕ_5, and ϕ_6. Combining equations (59) to (62), the information that I have learned from the reduced form coefficients can yield the following equations:

$$\phi_1 = \lambda_1(1+\beta) + \alpha + \beta\lambda_0 \tag{63}$$

$$\phi_2 = \lambda_2 \tag{64}$$

$$\phi_3 = \lambda_3(1+\beta) + \beta\lambda_2 \tag{65}$$

$$\phi_4 = \lambda_1 \tag{66}$$

$$\phi_5 = \lambda_2(1+\beta) + \alpha + \beta\lambda_0 \tag{67}$$

$$\phi_6 = \lambda_3(1+\beta) + \beta\lambda_1 \tag{68}$$

Equations (63) to (68) explicitly express the reduced-form coefficients as functions of underlying structural parameters. Solving this system yields $\lambda_1, \lambda_2, \lambda_3$, and β directly, together with the combination $\alpha + \beta\lambda_0$; under the condition $\lambda_1 \neq \lambda_2$, β is identified. The intercept λ_0 is then pinned by the normalization $\sum \theta_i = 0$, giving $\lambda_0 = -\lambda_1\overline{D_{i1}} - \lambda_2\overline{D_{i2}} - \lambda_3\overline{D_{i1}D_{i2}}$. This normalization serves two purposes: it separates the sector premium α from $\beta\lambda_0$, and it recovers the intercept needed to estimate the distribution of θ_i from equation (58).

9.4 Revealed Comparative Advantages

This appendix shows that the projection of θ_i can be interpreted as revealed comparative advantages, which contain rich empirical information in the context of Indonesia.

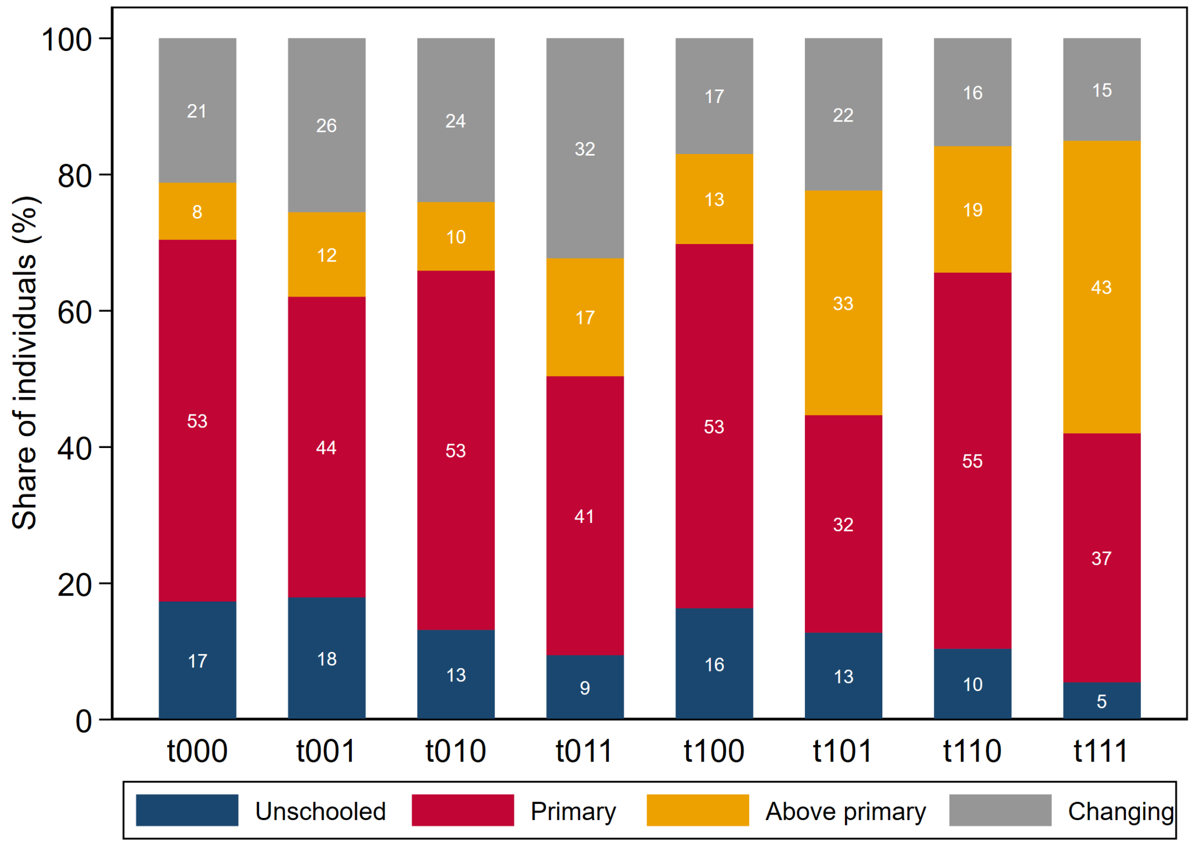
Although Chamberlain (1982, 1984) and Suri (2011) explicitly emphasize that equation (58) is primarily a technical device for eliminating correlation between θ_i and choice variable D_{it}, it can equivalently be interpreted as a regression of the latent comparative advantage θ_i on indicators of the choice trajectory (choices at each t and their interaction). In this formulation, the fitted values $\hat{\theta}_i$ represent the component of individual underlying comparative advantage explained by the trajectory, thereby providing an empirical measure of unobserved heterogeneity across groups.

Drawing loosely on the intuition of revealed preference theory (Samuelson, 1938, 1948), sectoral choices can be interpreted as revealing information about underlying comparative advantages. In this context, individuals implicitly conduct a cost–benefit analysis, where potential earnings in each sector are the benefits, and constraints such as schooling, time, and ability represent the costs. Over three waves, each agent's sequence of sectoral choices yields one of eight possible trajectories ($2^3 = 8$), reflecting how unobserved comparative advantages shape decisions. Although this analogy to revealed preference is only suggestive rather than a formal extension, the observed choice histories and their interactions provide an empirical basis for capturing latent abilities. I therefore refer to the fitted values from this procedure, $\hat{\theta}_i$, as revealed comparative advantages.

Figures 10 and 11 illustrate the composition of education levels and waged work across the eight trajectory groups, denoted by $t000, t001, t010, t011, t100, t101, t110, t111$, where 1 indicates non-agriculture and 0 indicates agriculture in a given period. For example, $t000$ corresponds to staying in agriculture in all three waves, while $t111$ corresponds to remaining in non-agriculture. Figure 10 shows clear contrasts: non-agricultural stayers ($t111$) are disproportionately drawn from individuals with education beyond primary school, whereas agricultural stayers ($t000$) contain a higher share of unschooled individuals. Sector switchers, by contrast, display larger shifts in educational attainment across waves. A parallel pattern emerges in Figure 11: switchers exhibit greater transitions between self-employment and waged work, while stayers tend to remain more stable in their employment types.

Figures 12 and 13 provide additional evidence that choice trajectories reveal information about underlying comparative advantages. A well-documented puzzle in the APG literature using IFLS data is that individuals who move from non-agriculture to agriculture appear to

Figure 10: Education Composition by Choice-Trajectory Group



experience substantial earnings losses, yet a proportion of workers choose to switch (Pulido and Świecki, 2019; Hamory et al., 2021). When earnings are instead examined by trajectory groups, as shown in Figure 12, the distributions of log earnings for all groups shift to the right over time, indicating earnings growth for each trajectory group. This perspective suggests that trajectory groups differ in their initial mean earnings, reflecting underlying abilities and costs across sectors. When outcomes are aggregated to sector-level averages, these initial differences are masked, creating the appearance of earnings losses that are not evident once trajectories are taken into account.

Figure 13 further shows that hours worked remain relatively stable for stayers but fluctuate for switchers. Taken together, these results suggest that trajectory groups capture systematic heterogeneity consistent with comparative advantages on which sectoral choices are made.

Figure 11: Waged-Work Status by Choice-Trajectory Group

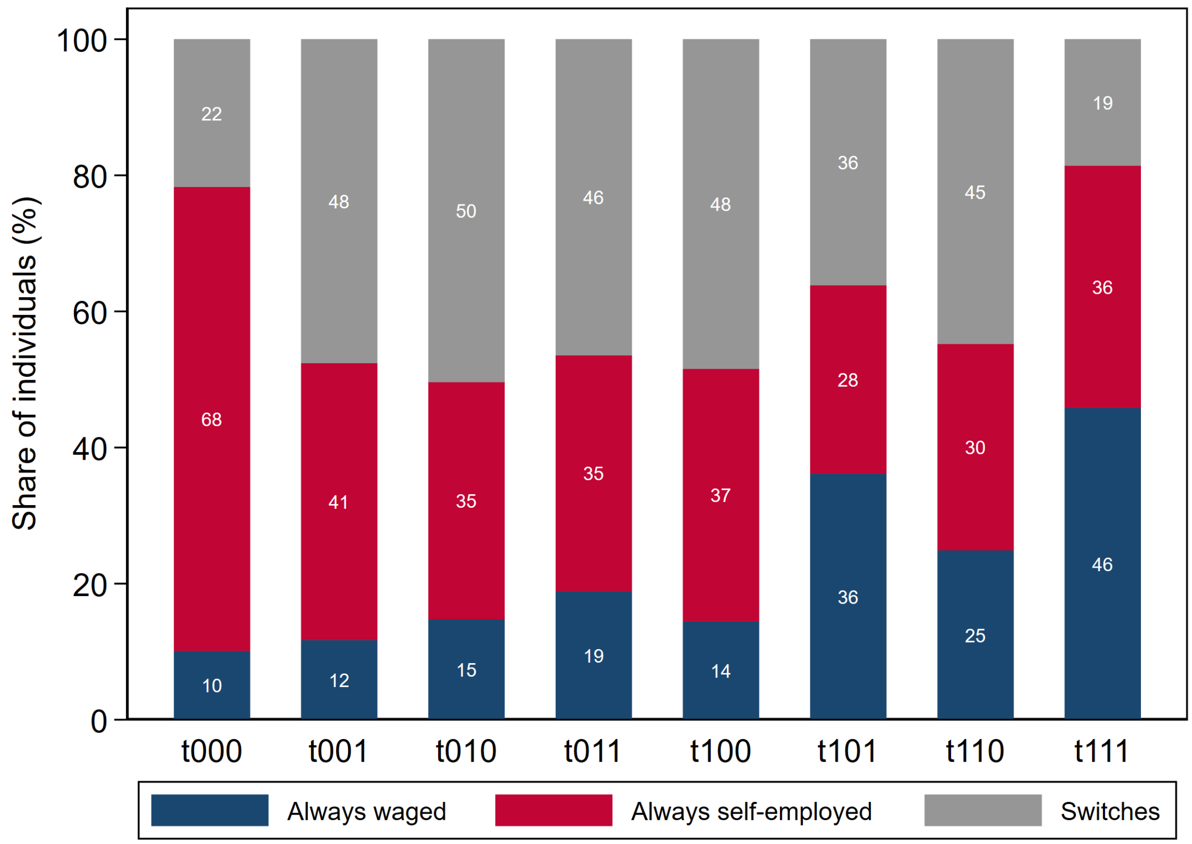


Figure 12: Log Earnings Distribution in 3 Waves by Choice Trajectory

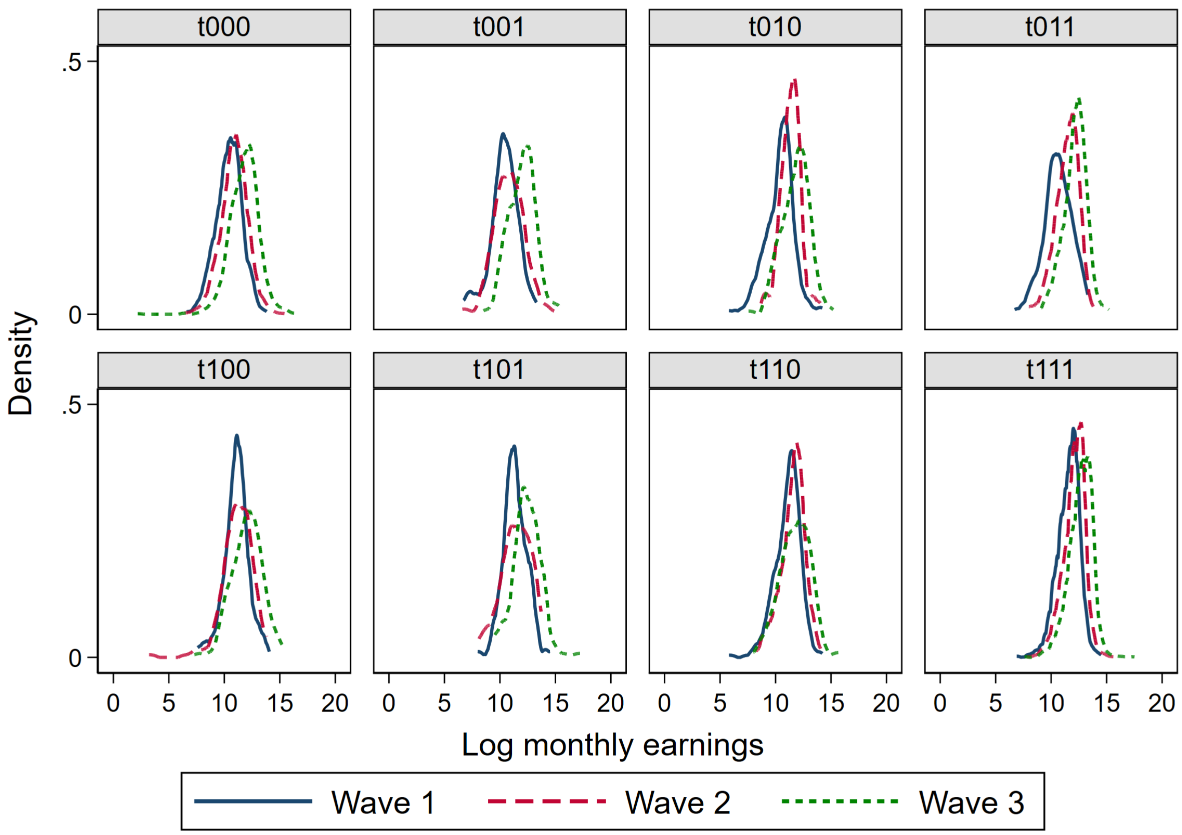
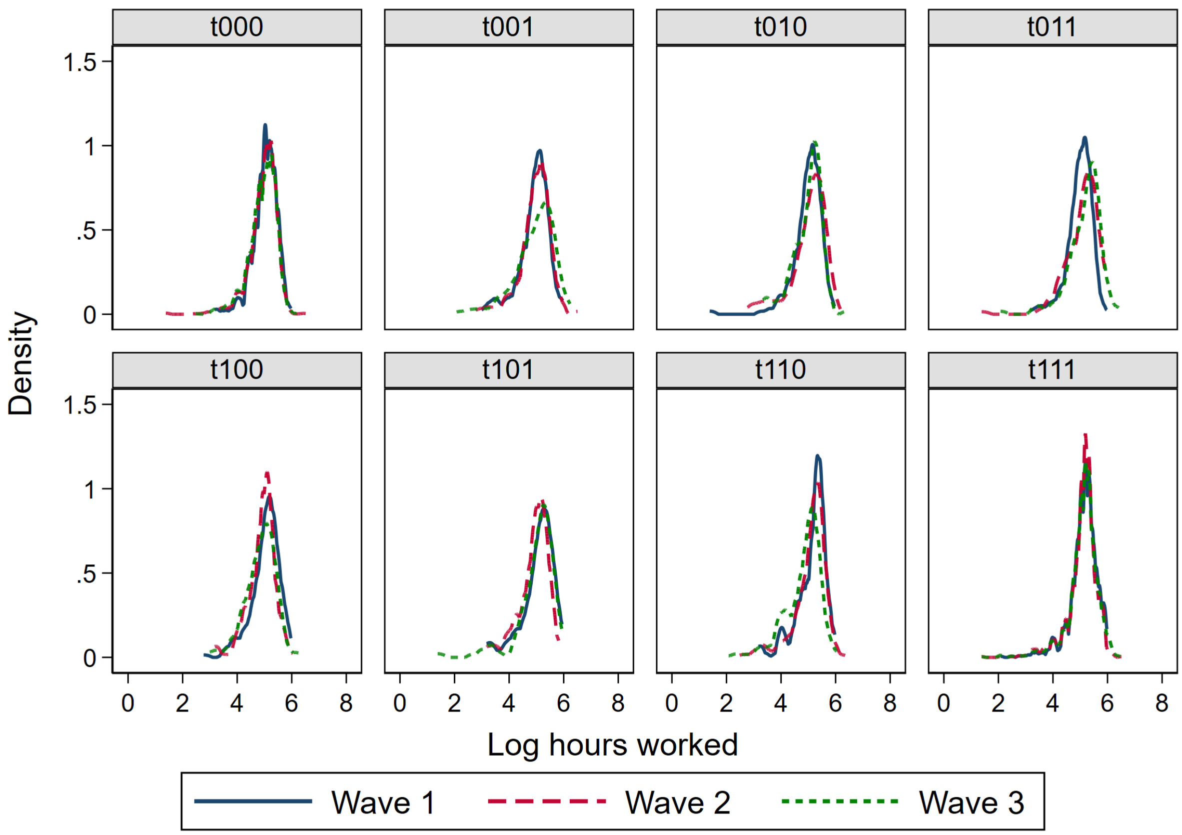


Figure 13: Log Hours Worked Distribution in 3 Waves by Choice Trajectory



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